

Computer Networks

A computer network is a system of interconnected devices that can communicate and share data with each other. Some examples of devices that can be part of a computer network are computers, printers, routers, switches, servers, smartphones, etc.

Some of the benefits of computer networks are:

- They enable resource sharing, such as files, printers, scanners, etc.
- They facilitate communication, such as email, chat, video conferencing, etc.
- They enhance security, such as encryption, authentication, firewall, etc.
- They improve performance, such as load balancing, caching, compression, etc.
- They support distributed applications, such as web, cloud, peer-to-peer, etc.

Some of the challenges of computer networks are:

- They require complex design, implementation, and management.
- They are vulnerable to failures, attacks, and congestion.
- They have to deal with heterogeneity, scalability, and compatibility issues.
- They have to comply with standards, protocols, and policies.

Some of the basic concepts of computer networks are:

- Network topology: The physical or logical arrangement of devices and links in a network.
- Network architecture: The structure and organization of network components and functions.
- Network protocol: The set of rules and conventions that govern the communication and interaction between network entities.
- Network layer: The level of abstraction that defines the functionality and scope of network services.
- Network model: The conceptual framework that describes the structure and behavior of a network system.

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

- A computer network is a collection of devices and systems that are connected by communication channels to facilitate the exchange of data and resources.
- The main components of a computer network are hosts, links, switches, routers, and protocols.
- Hosts are the end devices that generate and consume data, such as computers, smartphones, printers, etc.

- Links are the physical media that carry the data, such as wires, cables, optical fibers, radio waves, etc.
- Switches are devices that forward data within a local area network (LAN) based on the destination address of the data frames.
- Routers are devices that forward data between different networks based on the destination address of the data packets.
- Protocols are the rules and conventions that govern the communication between hosts and devices, such as TCP/IP, HTTP, FTP, etc.
- The main objectives of a computer network are to provide connectivity, reliability, security, scalability, and performance.
- Connectivity refers to the ability of hosts and devices to communicate with each other across the network.
- Reliability refers to the ability of the network to deliver data correctly and timely despite failures and errors.
- Security refers to the ability of the network to protect the data from unauthorized access, modification, or disclosure.
- Scalability refers to the ability of the network to accommodate the growth of hosts and devices without compromising the quality of service.
- Performance refers to the ability of the network to meet the requirements of the applications and users, such as throughput, latency, jitter, etc.
- The physical layer is the lowest layer of the network architecture that deals with the transmission and reception of raw bits over the physical medium.
- The main functions of the physical layer are to encode and decode the bits, modulate and demodulate the signals, and synchronize and multiplex the data streams.
- Encoding is the process of converting the bits into electrical or optical signals that can be transmitted over the link.
- Decoding is the process of converting the signals back into bits at the receiver end.
- Modulation is the process of varying the properties of a carrier wave (such as amplitude, frequency, or phase) according to the data bits to be transmitted.
- Demodulation is the process of extracting the data bits from the modulated carrier wave at the receiver end.
- Synchronization is the process of aligning the sender and receiver clocks to ensure the correct timing of the data transmission and reception.
- Multiplexing is the process of combining multiple data streams into one signal that can be transmitted over a shared medium.
- The main challenges of the physical layer are to overcome the noise, attenuation, interference, and distortion that affect the quality of the signal and the data transmission.

Introductory Concepts of Computer Networks

- A computer network is a system that connects numerous independent computers in order to share information (data) and resources.

- Computer networks help you to connect with multiple computers together to send and receive information.
- Computer networks follow protocols, which define how communications are sent and received.
- Computer networks consist of essential components, such as end devices, media, protocols, and networking devices.
- End devices are devices that send or receive the data in the network, such as computers, printers, and smartphones.
- Media provides connectivity between the end devices, such as cables, fiber optics, or wireless signals .
- Protocols enable communication between two or more end devices, such as TCP/IP, HTTP, and FTP.
- Networking devices are devices that facilitate the transmission and routing of data in the network, such as switches, routers, and wireless access points .
- Computer networks can be classified based on different criteria, such as topology, size, architecture, and functionality.
- Topology refers to the physical or logical arrangement of the network nodes and links, such as bus, ring, star, and mesh.
- Size refers to the geographical extent of the network, such as local area network (LAN), metropolitan area network (MAN), and wide area network (WAN).
- Architecture refers to the design and structure of the network, such as peer-to-peer (P2P), client-server, and cloud computing.
- Functionality refers to the purpose and scope of the network, such as intranet, extranet, and internet.

Goals and applications of networks and protocols

- A network is a collection of devices that can communicate with each other through a medium, such as wires, cables, radio waves, or optical fibers.
- A protocol is a set of rules and conventions that govern how devices exchange information over a network.
- The main goals of networks and protocols are:
 - To enable data sharing and collaboration among different users and applications.
 - To provide reliable and efficient transmission of data across different types of networks and devices.
 - To ensure security and privacy of data and users on the network.
 - To support scalability and interoperability of networks and devices.
- Some examples of applications of networks and protocols are:
 - The Internet, which is a global network of networks that uses the Internet Protocol (IP) and other protocols to connect billions of devices and users.
 - The World Wide Web, which is an application that uses the Hypertext Transfer Protocol (HTTP) and other protocols to deliver web

pages and multimedia content over the Internet.

- Email, which is an application that uses the Simple Mail Transfer Protocol (SMTP) and other protocols to send and receive messages over the Internet.
- Voice over IP (VoIP), which is an application that uses the Real-time Transport Protocol (RTP) and other protocols to transmit voice and video calls over the Internet.
- File Transfer Protocol (FTP), which is an application that uses the Transmission Control Protocol (TCP) and other protocols to transfer files between devices over the Internet.

Categories of networks in computer networks

A computer network is a system of interconnected devices that can communicate and share data, resources, and applications. Computer networks can be classified based on several criteria, such as the transmission medium, the network size, the topology, and organizational intent. Based on a geographical scale, the different types of networks are:

- **Personal area network (PAN):** This is the smallest and simplest type of network. PANs connect devices within the range of an individual and are no larger than about 10 meters. Examples of PANs are wireless headphones, Bluetooth speakers, smartwatches, etc.
- **Local area network (LAN):** This is the most frequently used network. A LAN is a computer network that connects devices in a limited physical area, such as a home, office, school, or laboratory. LANs use wired or wireless links to transmit data rapidly and reliably. Examples of LANs are Ethernet, Wi-Fi, etc.
- **Wide area network (WAN):** This is a computer network that spans a large geographical area, such as a country, continent, or the whole world. WANs use public or private communication channels, such as telephone lines, satellites, or optical fibers, to connect devices that are far apart. Examples of WANs are the Internet, cellular networks, etc.
- **Metropolitan area network (MAN):** This is a computer network that covers a large city or a metropolitan area. MANs are usually composed of several LANs that are interconnected by high-speed links, such as fiber optics or microwave. Examples of MANs are cable TV networks, municipal Wi-Fi networks, etc.
- **Campus area network (CAN):** This is a computer network that connects devices within a campus, such as a university, college, or corporate campus. CANs are similar to LANs, but they cover a larger area and may have multiple buildings or sites. Examples of CANs are academic networks, research networks, etc.
- **Storage area network (SAN):** This is a computer network that connects storage devices, such as hard disks, tape drives, or optical disks, to

servers or other devices. SANs provide fast and secure access to large amounts of data, and they are often used for backup, recovery, or data replication. Examples of SANs are Fibre Channel, iSCSI, etc.

- **Nanoscale network (NAN):** This is a computer network that enables communication between minuscule sensors and actuators, such as nanomachines, nanodevices, or nanorobots. NANs are used for biomedical, environmental, or industrial applications, and they may use molecular, electromagnetic, or acoustic signals. Examples of NANs are nanonetworks, nanosensors, etc.

Organization of the Internet

- The Internet is a global network of interconnected computers and devices that communicate using standardized protocols and procedures.
- The Internet is composed of many smaller networks, called internets, that are connected by routers and gateways.
- The Internet is organized at four levels: hardware, access, navigation, and community.
 - Hardware: The physical devices and infrastructure that enable data transmission and processing, such as computers, servers, cables, satellites, etc.
 - Access: The providers and services that allow users and organizations to connect to the Internet, such as Internet service providers (ISPs), broadband, dial-up, etc.
 - Navigation: The tools and systems that help users find and access information and resources on the Internet, such as domain names, IP addresses, web browsers, search engines, etc.
 - Community: The sites and platforms that host and facilitate online interactions and content creation, such as social media, blogs, forums, wikis, etc.
- The Internet is governed by a variety of organizations and entities that have different roles and responsibilities, such as:
 - Internet Society (ISOC): A non-profit organization that promotes the development, evolution, and use of the Internet for the benefit of all people.
 - Internet Corporation for Assigned Names and Numbers (ICANN): A non-profit organization that coordinates the global domain name system (DNS) and IP address allocation.
 - Internet Engineering Task Force (IETF): A voluntary group of network engineers and experts that develops and maintains the technical standards and protocols of the Internet.
 - World Wide Web Consortium (W3C): An international consortium of members that develops and maintains the standards and guidelines for the web.
 - Internet Governance Forum (IGF): A multi-stakeholder platform that facilitates dialogue and policy-making on Internet-related issues.

ISP

- ISP stands for **Internet Service Provider** .
- An ISP is a company that provides **access to the internet** to both personal and business customers .
- ISPs charge a **fee** for their services, which may vary depending on the speed, bandwidth, and type of connection.
- ISPs may also provide other services such as **email, domain registration, web hosting, and browser services** .
- ISPs can **host websites** for businesses and can also **build the websites** themselves.
- ISPs connect their customers to the internet through different methods, such as **cable, DSL, fiber-optic, satellite, or wireless**.
- ISPs are regulated by different laws and agencies depending on the country and region they operate in. Some of the issues that ISPs may face are **privacy, censorship, net neutrality, and competition**.

Network structure with reference to Computer Networks

- A computer network is a structure that makes available to a data processing user at one place some data processing function or service performed at another place.
- A computer network consists of two or more devices that can communicate with each other and share data, files, resources, or services.
- A computer network can be classified by its size, topology, architecture, or protocol.
- The size of a computer network refers to the geographical area it covers and the number of devices it connects. Some common network sizes are:
 - LAN (local area network): A LAN connects computers over a relatively short distance, such as within a building or a campus. A LAN typically uses wired media, such as Ethernet cables, to transmit data.
 - WLAN (wireless local area network): A WLAN is just like a LAN but connections between devices on the network are wireless, such as Wi-Fi or Bluetooth.
 - MAN (metropolitan area network): A MAN connects computers over a larger area, such as a city or a region. A MAN typically uses a combination of wired and wireless media, such as fiber-optic cables and microwave links, to transmit data.
 - WAN (wide area network): A WAN connects computers over a very large area, such as a country or a continent. A WAN typically uses public or leased telecommunication networks, such as the Internet, to transmit data.
- The topology of a computer network refers to the physical or logical arrangement of the devices and the links that connect them. Some common network topologies are:
 - Bus: A bus network has a single cable that connects all the devices

- on the network. Data is broadcasted to all the devices and only the intended recipient accepts it. A bus network is simple and cheap, but it has low performance and reliability.
- Star: A star network has a central device, such as a switch or a hub, that connects all the other devices on the network. Data is sent from one device to the central device, which then forwards it to the destination device. A star network is easy to install and manage, but it depends on the central device for its functionality.
 - Ring: A ring network has a circular arrangement of devices that are connected by a single cable. Data is transmitted in one direction around the ring, passing through each device until it reaches the destination. A ring network is efficient and reliable, but it has high latency and difficulty in adding or removing devices.
 - Mesh: A mesh network has multiple paths between any pair of devices on the network. Data is routed along the best available path, using algorithms or protocols. A mesh network is robust and fault-tolerant, but it has high cost and complexity.
 - Tree: A tree network has a hierarchical structure of devices that are connected by branches. Data is transmitted from the root device to the leaf devices, or vice versa, following the tree structure. A tree network is scalable and flexible, but it has high maintenance and security issues.
- The architecture of a computer network refers to the way network devices and services are structured and organized to serve the connectivity needs of client devices. Some common network architectures are:
 - Client-server: A client-server network has a central device, such as a server, that provides services to other devices, such as clients, on the network. Clients request services from the server, and the server responds to the requests. A client-server network is efficient and secure, but it requires a lot of resources and administration.
 - Peer-to-peer: A peer-to-peer network has no central device or authority, and each device on the network can act as both a client and a server. Devices communicate and share resources directly with each other, without any intermediaries. A peer-to-peer network is simple and decentralized, but it has low security and scalability.
 - Hybrid: A hybrid network combines the features of both client-server and peer-to-peer networks, depending on the situation and the application. A hybrid network can have both servers and peers, and devices can switch roles as needed. A hybrid network is flexible and adaptable, but it has high complexity and compatibility issues.
 - The protocol of a computer network refers to the set of rules and standards that govern how devices communicate and exchange data on the network. Some common network protocols are:
 - TCP/IP: TCP/IP is the predominant model for today's Internet structure and presents this standard layer configuration for communication links:

- * Network access layer: Defines how the data gets physically transferred.
- * Internet layer: Packages the data into understandable packets so it can be sent and received.
- * Transport layer: Allows

Network architecture with reference to Computer Networks

- Network architecture is the design of a computer network .
- It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used .
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols .
- Hardware refers to the devices that make up the network, such as computers, routers, switches, hubs, etc .
- Software refers to the programs that run on the network devices, such as operating systems, applications, network services, etc .
- Transmission media refers to the physical or wireless means of carrying data signals between network devices, such as copper wires, fiber optics, radio waves, etc .
- Network topology refers to the shape or layout of the network, such as bus, star, ring, mesh, etc .
- Communications protocols refer to the rules and standards that govern how network devices exchange data, such as TCP/IP, Ethernet, Wi-Fi, etc .
- Network architecture can be classified into different types based on the network's size and purpose, such as LAN, WAN, MAN, PAN, etc .
- LAN (local area network) connects computers over a relatively short distance, allowing them to share data, files, resources, etc .
- WAN (wide area network) connects computers over a large geographical area, such as across cities, countries, or continents .
- MAN (metropolitan area network) connects computers within a city or a metropolitan area, such as a campus, a company, or a government agency .
- PAN (personal area network) connects devices around a single person, such as a smartphone, a laptop, a smartwatch, etc .
- Network architecture can also be classified into different layers based on the level of abstraction and functionality, such as physical, data link, network, transport, session, presentation, and application layers .
- Physical layer defines the electrical and mechanical characteristics of the transmission media and the signals .
- Data link layer defines the methods of accessing the transmission media and the techniques of error detection and correction .
- Network layer defines the addressing and routing schemes of the network

and the protocols for packet switching and congestion control .

- Transport layer defines the mechanisms for reliable and efficient data transfer between end devices and the protocols for flow control and multiplexing .
- Session layer defines the methods of establishing, maintaining, and terminating sessions between network applications .
- Presentation layer defines the formats and standards for data representation and encryption .
- Application layer defines the interfaces and protocols for network services and applications, such as email, web, file transfer, etc .

Layering Principles with Reference to Network Architecture in Computer Networks

- Layering is a mechanism that divides a complex system into smaller and manageable parts, called layers, that interact with each other through well-defined interfaces .
- Layering allows for modularity, abstraction, encapsulation, and interoperability of network components and protocols.
- Layering also facilitates the design, implementation, testing, and maintenance of network systems, as well as the adaptation to new technologies and standards.
- One of the most widely used network models that employs layering is the Open Systems Interconnection (OSI) model, which consists of seven layers: physical, data link, network, transport, session, presentation, and application .
- The principles that were applied to arrive at the seven layers of the OSI model can be briefly summarized as follows:
 - A layer should be created where a different abstraction is needed.
 - Each layer should perform a well-defined function.
 - The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
 - The layer boundaries should be chosen to minimize the information flow across the interfaces.
 - The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.
- The OSI model defines the services, functions, and protocols for each layer, as well as the data units and formats that are exchanged between adjacent layers.
- The data units are called protocol data units (PDUs), which consist of headers, trailers, and payloads. Headers and trailers contain control information, such as source and destination addresses, sequence numbers, error detection codes, etc. Payloads contain the actual data or information.
- The PDUs are encapsulated and decapsulated as they traverse the layers. For example, when data is sent from the application layer to the physical

layer, each layer adds its own header and trailer to the PDU received from the upper layer, forming a new PDU. When data is received from the physical layer to the application layer, each layer removes the header and trailer added by the corresponding layer at the sender side, and passes the remaining PDU to the upper layer.

- The OSI model provides a conceptual framework for understanding the network architecture and communication process, but it is not a practical implementation. In reality, most network systems use a simplified version of the OSI model, such as the TCP/IP model, which has four layers: application, transport, internet, and network access.

Services in Networks Architecture in Computer Networks

- Services in networks architecture are the **functions** or **operations** that each layer of the network model provides to the layer above it or below it.
- Services are **defined** by a set of **primitives** or **commands** that specify the **parameters** and **results** of the service.
- Services are **implemented** by using **protocols** or **rules** that govern the **interaction** and **communication** between the entities at each layer.
- Services can be **classified** into two types: **connection-oriented** and **connectionless**.
 - Connection-oriented services require the establishment of a **logical connection** between the sender and the receiver before any data can be transferred. The connection is **terminated** after the data transfer is complete. An example of a connection-oriented service is **TCP**.
 - Connectionless services do not require any prior **arrangement** or **coordination** between the sender and the receiver. The sender can send data to the receiver at any time, without any **acknowledgment** or **confirmation**. An example of a connectionless service is **UDP**.
- Services can also be **categorized** into two levels: **peer-to-peer** and **end-to-end**.
 - Peer-to-peer services are the services that are **exchanged** between the **same layer** of different network devices. For example, the **data link layer** of one device can exchange **frames** with the data link layer of another device using a peer-to-peer service.
 - End-to-end services are the services that are **delivered** from the **source** to the **destination** across multiple network devices and layers. For example, the **application layer** of one device can send **messages** to the application layer of another device using an end-to-end service.
- Some examples of services in networks architecture are:
 - **DHCP** (Dynamic Host Configuration Protocol): A service that **assigns** IP addresses and other network configuration parameters to devices on a network.
 - **DNS** (Domain Name System): A service that **translates** domain names into IP addresses and vice versa.

- **FTP** (File Transfer Protocol): A service that **transfers** files between devices on a network.
- **HTTP** (Hypertext Transfer Protocol): A service that **requests** and **delivers** web pages and other resources on the internet.
- **SMTP** (Simple Mail Transfer Protocol): A service that **sends** and **receives** email messages on a network.

Protocols and Standards in Networks Architecture in Computer Networks

- Network protocols are the rules and procedures that define how devices communicate and exchange data over a network. They specify the format, timing, sequence, and error control of the data transmission.
- Network standards are the agreed-upon conventions and guidelines that ensure interoperability and compatibility among different devices, vendors, and technologies in a network. They are developed and maintained by various organizations and committees, such as IEEE, IETF, ISO, etc.
- Network architecture is the design and structure of a network, including its components, protocols, standards, and topology. It describes how the network functions, operates, and performs.
- Some of the common network protocols and standards are:
 - TCP/IP: The Transmission Control Protocol/Internet Protocol is the most widely used protocol suite in the Internet and other networks. It consists of four layers: application, transport, internet, and network interface. It provides reliable, connection-oriented, and end-to-end data delivery.
 - UDP: The User Datagram Protocol is a connectionless and unreliable protocol that operates at the transport layer. It provides fast and efficient data transmission for applications that do not require reliability or error control, such as streaming media, online gaming, etc.
 - IP: The Internet Protocol is the core protocol of the internet layer. It is responsible for routing and addressing packets across different networks. It uses a 32-bit or 128-bit address to identify each device on the network.
 - ARP: The Address Resolution Protocol is a protocol that maps an IP address to a physical address, such as a MAC address. It operates at the network interface layer and uses broadcast messages to request and reply the address information.
 - DHCP: The Dynamic Host Configuration Protocol is a protocol that assigns IP addresses and other network configuration parameters to

devices dynamically. It operates at the application layer and uses a client-server model to exchange messages.

- FTP: The File Transfer Protocol is a protocol that enables the transfer of files between a client and a server over a network. It operates at the application layer and uses TCP for reliable data transmission. It supports both binary and ASCII modes of file transfer.
- DNS: The Domain Name System is a protocol that translates domain names to IP addresses and vice versa. It operates at the application layer and uses a hierarchical and distributed database to store the name and address information. It uses UDP for query and response messages.
- HTTP: The Hypertext Transfer Protocol is a protocol that defines how web browsers and web servers communicate and exchange web resources, such as HTML documents, images, etc. It operates at the application layer and uses TCP for reliable data transmission. It supports various methods, such as GET, POST, PUT, DELETE, etc.
- SMTP: The Simple Mail Transfer Protocol is a protocol that enables the sending and receiving of email messages over a network. It operates at the application layer and uses TCP for reliable data transmission. It supports various commands, such as HELO, MAIL, RCPT, DATA, QUIT, etc.

The OSI reference model in in Computer Networks

The OSI reference model is a conceptual framework that defines how different devices and applications communicate over a network. It stands for Open System Interconnection and was developed by the International Organization for Standardization (ISO) in 1984. The model consists of seven layers, each of which performs a specific function and interacts with the adjacent layers. The purpose of the model is to facilitate interoperability and standardization among different network systems and protocols.

The seven layers of the OSI model are:

- **Layer 1: Physical layer.** This layer deals with the physical transmission of data over the network media, such as cables, wires, or wireless signals. It defines the electrical, mechanical, and functional characteristics of the network devices and interfaces, such as voltage levels, connectors, and encoding schemes. It also handles the conversion of digital data into analog signals or vice versa.
- **Layer 2: Data link layer.** This layer provides reliable and error-free data transfer between two devices on the same network segment, such

as a switch and a host. It defines the format and structure of the data frames, as well as the addressing and access control methods, such as MAC addresses and Ethernet protocols. It also performs error detection and correction, flow control, and synchronization of the data frames.

- **Layer 3: Network layer.** This layer provides logical addressing and routing of data packets across different network segments or domains, such as routers and gateways. It defines the network addresses, such as IP addresses and subnet masks, and the protocols for routing and forwarding, such as IP, ICMP, and ARP. It also performs fragmentation and reassembly, congestion control, and quality of service (QoS) of the data packets.
- **Layer 4: Transport layer.** This layer provides end-to-end data delivery and reliability between two applications on different hosts, such as a web server and a web browser. It defines the port numbers, such as TCP and UDP ports, and the protocols for establishing, maintaining, and terminating connections, such as TCP, UDP, and SCTP. It also performs segmentation and reassembly, error detection and recovery, flow control, and multiplexing of the data segments.
- **Layer 5: Session layer.** This layer provides the management and coordination of the communication sessions between two applications, such as a file transfer or a video conference. It defines the session identifiers, such as cookies and tokens, and the protocols for initiating, controlling, and terminating sessions, such as RPC, NFS, and SQL. It also performs authentication, authorization, encryption, and synchronization of the data exchange.
- **Layer 6: Presentation layer.** This layer provides the representation and transformation of the data formats and syntaxes between different applications, such as ASCII and Unicode. It defines the encoding, compression, encryption, and decryption methods, such as MIME, JPEG, and SSL. It also performs data translation, conversion, and interpretation of the data exchange.
- **Layer 7: Application layer.** This layer provides the interface and functionality of the network services and applications, such as email, web, and FTP. It defines the application protocols, such as HTTP, SMTP, and FTP, and the user interfaces, such as GUI and CLI. It also performs data validation, formatting, and processing of the data exchange.

The OSI model is a useful tool for understanding and designing computer networks, as it provides a common language and framework for describing the functions and interactions of different network components and protocols. By following the OSI model, network systems and applications can achieve compatibility and interoperability across different platforms and vendors.

TCP/IP protocol suite in in Computer Networks

- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices

on the internet or other computer networks .

- TCP/IP is also known as the internet protocol suite, as it defines the basic rules and methods for data transmission and addressing over the internet .
- TCP/IP is based on a four-layer model, which consists of the following layers :
 - Application layer: This layer provides the interface and services for user applications, such as web browsers, email clients, file transfer programs, etc. Some of the protocols in this layer are HTTP, SMTP, FTP, DNS, etc.
 - Transport layer: This layer provides reliable and efficient data delivery between the source and destination hosts, using protocols such as TCP and UDP. TCP ensures reliability, flow control, congestion control, and error detection and correction, while UDP provides fast and simple data transfer without reliability guarantees.
 - Internet layer: This layer provides the logical addressing and routing of data packets across different networks, using protocols such as IP, ICMP, ARP, etc. IP assigns a unique address to each host and device on the internet, and routes the packets based on the destination address. ICMP provides error and control messages, while ARP resolves the physical address of a host from its IP address.
 - Network access layer: This layer provides the physical connection and transmission of data between the hosts and the network devices, such as routers, switches, hubs, etc. This layer uses protocols such as Ethernet, Wi-Fi, PPP, etc. to encode and decode the data into electrical signals or radio waves, and to access the shared medium.
- TCP/IP is the most widely used protocol suite in the world, as it enables the interoperability and scalability of the internet and other computer networks . It is also flexible and adaptable, as it can support different types of networks, applications, and devices. TCP/IP is constantly evolving and improving, as new protocols and standards are developed and implemented.

Network devices in Computer Networks

- Network devices are hardware components that connect computers and other devices to a network. They enable data transmission and communication among different network nodes.
- Some common network devices are:
 - **Router:** A device that forwards packets between different networks based on their destination IP addresses. Routers can also perform network address translation (NAT), firewall, and routing protocols functions.
 - **Switch:** A device that connects multiple devices on the same network segment and forwards packets based on their destination MAC

addresses. Switches can also perform VLAN, QoS, and spanning tree functions.

- **Hub:** A device that connects multiple devices on the same network segment and broadcasts packets to all connected devices. Hubs are less efficient and secure than switches, as they create more collisions and do not filter traffic.
- **Bridge:** A device that connects two network segments and forwards packets based on their destination MAC addresses. Bridges can also perform filtering, learning, and flooding functions.
- **Repeater:** A device that amplifies and regenerates signals on a network segment. Repeaters can extend the range of a network, but they do not filter or forward traffic.
- **Modem:** A device that modulates and demodulates analog signals into digital signals and vice versa. Modems can enable data transmission over telephone lines, cable lines, or wireless networks.
- **NIC:** A device that provides a physical interface between a computer and a network. NICs can also perform encoding, decoding, framing, and error detection functions.
- **Firewall:** A device that monitors and controls incoming and outgoing network traffic based on predefined rules. Firewalls can also perform packet filtering, stateful inspection, proxy, and NAT functions.
- **Access point:** A device that allows wireless devices to connect to a wired network. Access points can also perform encryption, authentication, and roaming functions.

Network Components in Computer Networks

Network components are the devices and software that enable communication and data exchange between computers and other networked devices. Some of the major network components are:

- **Network Interface Card (NIC):** A NIC is a hardware device that allows a computer to connect to a network. It provides a physical interface for sending and receiving data over a network medium, such as a cable or wireless signal. A NIC also has a unique address, called a MAC address, that identifies the device on the network .
- **Hub:** A hub is a device that splits a network connection into multiple ports, allowing multiple computers to share the same physical link. A hub operates at the physical layer of the network, and does not perform any filtering or routing of data. A hub broadcasts all the data it receives to all the connected devices, regardless of their destination .
- **Switch:** A switch is a device that connects multiple devices on a network and forwards data only to the intended recipient. A switch operates at the data link layer of the network, and uses MAC addresses to identify the devices on the network. A switch can improve network performance

and security by reducing collisions and congestion .

- **Cables and connectors:** Cables and connectors are the physical components that link devices on a network. They provide the medium for transmitting electrical or optical signals between devices. There are different types of cables and connectors, such as coaxial, twisted pair, fiber optic, RJ-45, BNC, etc. The choice of cable and connector depends on the network speed, distance, and environment .
- **Router:** A router is a device that connects multiple networks and forwards data packets between them. A router operates at the network layer of the network, and uses IP addresses to identify the source and destination of data packets. A router can perform various functions, such as routing protocols, network address translation, firewall, etc. A router can enable communication between different network types, such as LAN, WAN, WLAN, etc .
- **Modem:** A modem is a device that modulates and demodulates signals between analog and digital formats. A modem enables a computer to communicate over a telephone line, cable, or wireless network. A modem can convert digital data from a computer into analog signals that can be transmitted over a phone line, and vice versa. A modem can also provide other functions, such as error correction, compression, encryption, etc .
- **Server:** A server is a computer that provides services to other computers on a network. A server can host various types of applications, such as web, email, database, file, etc. A server can process requests from clients and send responses back to them. A server can also store and manage data and resources for the network .
- **Client:** A client is a computer that requests services from a server on a network. A client can run various types of software, such as web browsers, email clients, database clients, etc. A client can send requests to a server and receive responses from it. A client can also access and use data and resources provided by the server .
- **Firewall:** A firewall is a device or software that monitors and controls the incoming and outgoing network traffic. A firewall can filter the traffic based on predefined rules, such as IP addresses, ports, protocols, etc. A firewall can prevent unauthorized access, block malicious attacks, and enforce network policies .
- **Access point:** An access point is a device that allows wireless devices to connect to a wired network. An access point acts as a bridge between the wireless and wired networks, and provides a common communication channel for the wireless devices. An access point can also perform functions such as authentication, encryption, and signal amplification .

Physical Layer in Computer Networks

- The physical layer is the lowest and first layer of the OSI model of computer networking.
- The physical layer provides an electrical, mechanical, and procedural in-

terface to the transmission medium that connects network nodes.

- The physical layer defines the means of transmitting a stream of raw bits over a physical data link.
- The physical layer performs the following functions:
 - It maintains the data rate (how many bits a sender can send per second).
 - It performs synchronization of bits (how the sender and receiver agree on when a bit starts and ends).
 - It helps in transmission medium decision (direction of data transfer, such as simplex, half-duplex, or full-duplex).
 - It helps in physical topology decision (shape of the network, such as mesh, star, bus, or ring).
 - It encodes and decodes the data into signals (such as analog or digital).
 - It transmits and receives the signals over the transmission medium (such as copper wire, optical fiber, or wireless).
- The physical layer can be implemented through a great number of different hardware technologies with widely varying characteristics.
- Some examples of physical layer standards are Ethernet, Wi-Fi, Bluetooth, USB, and RS-232.

Network topology design in Computer Networks

- Network topology is the arrangement of the nodes and links in a network, both physically and logically.
- Physical topology refers to the actual layout of the cables and devices, while logical topology refers to how data flows between the nodes.
- Network topology affects the performance, reliability, scalability, and security of a network.
- There are different types of network topologies, such as bus, ring, star, mesh, tree, and hybrid.
- Bus topology: All the nodes are connected to a single cable, called a bus or backbone. Data is transmitted in one direction along the bus. This topology is simple and cheap, but prone to collisions and faults.
- Ring topology: All the nodes are connected in a closed loop, forming a ring. Data is transmitted in one direction around the ring. This topology is simple and reliable, but has low bandwidth and high latency.
- Star topology: All the nodes are connected to a central device, called a hub or switch. Data is transmitted from the source node to the hub, and then to the destination node. This topology is easy to install and manage, but requires more cables and devices, and the hub is a single point of failure.

- **Mesh topology:** All the nodes are connected to each other, forming a mesh. Data is transmitted along the shortest path between the source and destination nodes. This topology is robust and fault-tolerant, but complex and expensive.
- **Tree topology:** All the nodes are connected in a hierarchical structure, forming a tree. Data is transmitted from the root node to the branch nodes, and then to the leaf nodes. This topology is scalable and flexible, but requires more cables and devices, and the root node is a single point of failure.
- **Hybrid topology:** A combination of two or more topologies, forming a hybrid. Data is transmitted according to the rules of each topology. This topology is adaptable and customizable, but complex and difficult to troubleshoot.
- **Network topology design** is the process of choosing the most suitable topology for a given network, based on the following factors:
 - The size and scope of the network
 - The number and type of nodes and links
 - The bandwidth and latency requirements
 - The cost and availability of resources
 - The reliability and security needs
 - The scalability and flexibility options
- **Network topology design** involves the following steps:
 - Analyzing the network requirements and objectives
 - Evaluating the advantages and disadvantages of each topology
 - Comparing the performance and cost of each topology
 - Selecting the best topology for the network
 - Implementing and testing the network topology
 - Monitoring and maintaining the network topology

Types of connections in Computer Networks

A connection in a computer network is a link between two or more devices that allows them to communicate and share data, resources, and applications. There are different types of connections in computer networks, depending on the topology, size, and function of the network. Some of the common types of connections are:

- **Point-to-point connection:** This type of connection allows one device to communicate with one other device. For example, two phones may pair with each other using Bluetooth to exchange files or make calls. A point-to-point connection is usually established by a cable or a wireless signal. A point-to-point connection is simple, secure, and efficient, but it can only support two devices at a time.

- **Broadcast/multicast connection:** This type of connection allows a device to send one message out to the network and have copies of that message delivered to multiple recipients. For example, a radio station may broadcast its program to many listeners, or a video conference may multicast its stream to many participants. A broadcast/multicast connection is usually established by a wireless signal or a shared medium, such as a coaxial cable. A broadcast/multicast connection is useful for mass communication, but it can cause congestion and interference in the network.
- **Multipoint connection:** This type of connection allows one device to connect and deliver messages to multiple devices in parallel. For example, a hub may connect several computers in a local area network (LAN), or a router may connect several LANs in a wide area network (WAN). A multipoint connection is usually established by a switch or a router, which can direct the traffic to the appropriate destination. A multipoint connection is flexible and scalable, but it can introduce delays and errors in the transmission.
- **LAN (local area network) connection:** This type of connection connects computers over a relatively short distance, allowing them to share data, files, and resources. For example, a LAN may connect all the computers in an office building, school, or hospital. A LAN connection is usually established by a cable, such as an Ethernet cable, or a wireless signal, such as Wi-Fi. A LAN connection is fast, reliable, and secure, but it can only cover a limited area .
- **MAN (metropolitan area network) connection:** This type of connection connects computers over a larger area, such as a city or a campus. For example, a MAN may connect several LANs in a university or a corporation. A MAN connection is usually established by a fiber optic cable, which can transmit data at high speeds and long distances. A MAN connection is faster and more reliable than a WAN connection, but it is more expensive and difficult to maintain.
- **PAN (personal area network) connection:** This type of connection connects devices that are close to a person, such as a smartphone, a laptop, a smartwatch, or a headset. For example, a PAN may connect a user's devices using Bluetooth, infrared, or USB. A PAN connection is usually established by a wireless signal or a short cable, which can provide convenience and mobility. A PAN connection is low-cost and easy to set up, but it has a limited range and bandwidth.
- **WAN (wide area network) connection:** This type of connection connects computers over a very large area, such as a country or a continent. For example, a WAN may connect several MANs or LANs across the world. A WAN connection is usually established by a satellite, a microwave, or a leased line, which can span long distances and cross geographical bound-

aries. A WAN connection is the most extensive and complex type of connection, but it is also the slowest and least reliable .

Transmission media in Computer Networks

- Transmission media is the physical medium that carries data from one device to another in a computer network .
- Transmission media can be classified into two types: guided media and unguided media.
- Guided media, also known as wired or bounded media, uses cables or wires to transmit data. Examples of guided media are twisted pair, coaxial cable, and optical fiber.
- Unguided media, also known as wireless or unbounded media, uses electromagnetic waves or signals to transmit data. Examples of unguided media are radio waves, microwaves, infrared, and visible light.
- The choice of transmission media depends on several factors, such as bandwidth, cost, distance, reliability, security, and interference .
- Transmission media is an essential component of network hardware, along with user devices, routers, servers, and gateways.
- Transmission media is also influenced by the network architecture, which is the design and structure of a network. Network architecture includes hardware, software, network topology, and communication protocols.

Signal transmission and encoding in Computer Networks

- Signal transmission is the process of sending and receiving data over a physical medium, such as a wire, a fiber optic cable, or a wireless channel.
- Encoding is the process of converting data into a format that can be transmitted as a signal, such as a voltage level, a light pulse, or a radio wave.
- The purpose of encoding is to ensure reliable and efficient transmission of data, by minimizing errors, interference, and bandwidth requirements.
- There are different types of encoding, depending on the nature of the data and the signal. Some common types are:
 - Digital-to-digital encoding: This is the conversion of binary data (0s and 1s) into a digital signal, such as a series of voltage pulses. This is also known as line coding. Examples of digital-to-digital encoding schemes are NRZ, Manchester, and 4B/5B.
 - Analog-to-digital encoding: This is the conversion of analog data, such as voice or video, into a digital signal, such as a sequence of bits. This involves sampling, quantization, and encoding. Examples of analog-to-digital encoding schemes are PCM, DM, and DPCM.
 - Digital-to-analog encoding: This is the conversion of digital data into an analog signal, such as a sine wave. This involves modulation,

which is the process of varying one or more properties of the analog signal, such as amplitude, frequency, or phase, according to the digital data. Examples of digital-to-analog encoding schemes are ASK, FSK, and PSK.

- Analog-to-analog encoding: This is the conversion of analog data into another analog signal, such as a different frequency band. This also involves modulation, which is the process of varying one or more properties of the analog signal, such as amplitude, frequency, or phase, according to the analog data. Examples of analog-to-analog encoding schemes are AM, FM, and PM.

Network performance and transmission impairments in Computer Networks

- Network performance is the measure of how well a network can deliver data and services to its users. It can be evaluated by various metrics, such as throughput, delay, jitter, packet loss, availability, reliability, etc.
- Transmission impairments are the factors that degrade the quality of a signal as it travels through a medium. They can cause distortion, attenuation, noise, interference, etc. Some common transmission impairments are:
 - Attenuation: The loss of signal strength due to the resistance of the medium, the distance traveled, or the presence of obstacles. It can be measured in decibels (dB) and can be compensated by amplifiers or repeaters.
 - Distortion: The alteration of the shape or frequency of a signal due to the characteristics of the medium or the devices. It can cause errors or misinterpretation of the data. It can be reduced by using filters or equalizers.
 - Noise: The unwanted or random signals that are added to the original signal due to the thermal agitation of electrons, electromagnetic radiation, or other sources. It can affect the signal-to-noise ratio (SNR) and the bit error rate (BER) of the transmission. It can be minimized by using shielding, modulation, or error correction techniques.
 - Interference: The overlapping or superposition of signals from different sources that use the same frequency or channel. It can cause collisions or congestion in the network. It can be avoided by using orthogonal or non-overlapping channels, frequency division multiplexing (FDM), or code division multiple access (CDMA).
 - Crosstalk: The coupling or leakage of signals from one wire or channel to another due to the electromagnetic induction or capacitance. It can cause unwanted signals or noise in the adjacent wires or channels. It can be reduced by using twisted pair cables, coaxial cables, or optical fibers.

Switching techniques and multiplexing in Computer Networks

- Switching techniques are methods of connecting multiple devices in a network and enabling them to communicate with each other efficiently.
- There are three main types of switching techniques: circuit switching, message switching, and packet switching.
- Circuit switching is a technique where a dedicated communication path is established between two nodes before data transmission. The path remains active until the communication is terminated. Circuit switching is suitable for real-time applications that require constant bandwidth and low delay, such as voice and video calls .
- Message switching is a technique where the whole message is treated as a data unit and stored and forwarded by intermediate nodes until it reaches the destination. Message switching does not require a dedicated path, but it introduces variable delay and overhead. Message switching is suitable for applications that can tolerate delay and do not require synchronization, such as email and file transfer .
- Packet switching is a technique where the message is broken down into smaller chunks called packets, which are transmitted independently and reassembled at the destination. Packet switching also does not require a dedicated path, but it allows efficient sharing of network resources and dynamic routing. Packet switching is suitable for applications that require high throughput and reliability, such as web browsing and streaming .
- Multiplexing is a technique of combining multiple signals into one signal over a shared medium. Multiplexing allows the utilization of the available bandwidth and reduces the cost of transmission. There are different types of multiplexing, such as frequency division multiplexing (FDM), time division multiplexing (TDM), and statistical multiplexing .
- Frequency division multiplexing (FDM) is a technique where the available bandwidth is divided into multiple frequency bands, and each signal is modulated by a different carrier frequency. FDM allows simultaneous transmission of multiple signals without interference, but it requires filters and guard bands to separate the signals. FDM is suitable for analog signals, such as radio and television .
- Time division multiplexing (TDM) is a technique where the available time is divided into multiple time slots, and each signal is assigned a different time slot. TDM allows sequential transmission of multiple signals without interference, but it requires synchronization and buffering to align the signals. TDM is suitable for digital signals, such as telephone and data .
- Statistical multiplexing is a technique where the available bandwidth is dynamically allocated to the signals based on their demand and priority. Statistical multiplexing does not require fixed division of bandwidth or time, but it introduces variable delay and congestion. Statistical multiplexing is suitable for packet switching networks, where the traffic is bursty and unpredictable .
- Switching and multiplexing are complementary techniques that enable efficient and reliable communication in computer networks. Switching connects multiple devices and routes data, while multiplexing shares net-

work resources and reduces cost. Together, they form the basis of modern network architectures and protocols.

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

- The link layer is the lowest layer in the TCP/IP model. It is responsible for sending and receiving frames between nodes on the same physical link.
- A frame is a unit of data that contains a header, a payload, and a trailer. The header contains information such as the source and destination addresses, the type of data, and the error detection code. The payload is the actual data to be transmitted. The trailer contains a special sequence of bits that marks the end of the frame.
- The link layer has two main functions: framing and error control. Framing is the process of dividing the data stream into frames and adding headers and trailers. Error control is the process of detecting and correcting errors that may occur during transmission.
- The link layer can use different techniques for error control, such as parity bits, checksums, cyclic redundancy checks (CRCs), and automatic repeat request (ARQ). Parity bits are extra bits that are added to the data to make the number of 1s even or odd. Checksums are values that are computed from the data and appended to the frame. CRCs are more complex checksums that use polynomial division. ARQ is a protocol that requires the sender to wait for an acknowledgment from the receiver before sending the next frame. If the acknowledgment is not received within a certain time, the sender retransmits the frame.
- The link layer can also use different techniques for medium access control (MAC), which is the process of coordinating the access of multiple nodes to a shared medium, such as a cable or a wireless channel. MAC techniques can be classified into two categories: deterministic and probabilistic. Deterministic techniques use a predefined order or schedule to determine which node can transmit at a given time. Probabilistic techniques use randomization or contention to decide which node can transmit at a given time.
- Some examples of deterministic MAC techniques are time division multiple access (TDMA), frequency division multiple access (FDMA), and code division multiple access (CDMA). TDMA divides the time into slots and assigns each slot to a node. FDMA divides the frequency into channels and assigns each channel to a node. CDMA assigns a unique code to each node and allows them to transmit simultaneously using different codes.
- Some examples of probabilistic MAC techniques are carrier sense multiple access (CSMA), carrier sense multiple access with collision detection (CSMA/CD), and carrier sense multiple access with collision avoidance (CSMA/CA). CSMA requires the node to sense the medium before transmitting and wait if the medium is busy. CSMA/CD requires the node to sense the medium during transmission and abort if a collision is de-

tected. CSMA/CA requires the node to exchange control frames with the intended receiver before transmitting and avoid collisions by using a backoff algorithm.

- A local area network (LAN) is a network that connects devices within a small geographic area, such as a building or a campus. LANs typically use a single link layer technology and a single MAC technique. LANs can be classified into two types: wired and wireless. Wired LANs use cables, such as twisted pair, coaxial, or fiber optic, to connect the devices. Wireless LANs use radio waves, such as infrared, microwave, or radio frequency, to connect the devices.
- Some examples of wired LAN technologies are Ethernet, token ring, and fiber distributed data interface (FDDI). Ethernet is the most widely used LAN technology. It uses CSMA/CD as the MAC technique and supports various data rates, such as 10 Mbps, 100 Mbps, 1 Gbps, and 10 Gbps. Token ring is a LAN technology that uses a ring topology and a token-passing protocol as the MAC technique. It supports a data rate of 4 Mbps or 16 Mbps. FDDI is a LAN technology that uses a dual ring topology and a token-passing protocol as the MAC technique. It supports a data rate of 100 Mbps and uses fiber optic cables.
- Some examples of wireless LAN technologies are Wi-Fi, Bluetooth, and ZigBee. Wi-Fi is the most widely used wireless LAN technology. It uses CSMA/CA as the MAC technique and supports various standards, such as 802.11a, 802.11b, 802.11g, 802.11n, and 802.11ac. These standards differ in the frequency band, the modulation scheme, the channel width, and the data rate. Bluetooth is a wireless LAN technology that uses a frequency hopping spread spectrum (FHSS) technique and supports various versions, such as Bluetooth 1.0, Bluetooth 2.0, Bluetooth 3.0, and Bluetooth 4.0.

Link layer in Computer Networks

- The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet.
- The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to.
- The link layer is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer may also provide the means to detect and possibly correct errors that can occur in the physical layer.
- The link layer is concerned with local delivery of frames between nodes on the same level of the network.
- The link layer can be divided into two sublayers: data link control and multiple access resolution/protocol.
- Data link control is the sublayer that handles framing, addressing, error control, and flow control.
- Multiple access resolution/protocol is the sublayer that handles the access of multiple nodes to a shared medium, such as a wireless channel or a bus

network.

- Some examples of multiple access protocols are ALOHA, CSMA, CSMA/CA, and CSMA/CD.

Framing in link layer in Computer Networks

- Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver.
- Frames are the result of the final layer of encapsulation before the data is transmitted over the physical layer. A frame consists of a link layer header followed by a packet.
- The link layer header contains information such as source and destination addresses, error-checking codes, and protocol identifiers.
- Framing uses frames to send or receive data. The data link layer receives packets from the network layer and converts them into frames.
- Framing also involves dividing the data stream into frames of fixed or variable size, adding delimiters to mark the boundaries of each frame, and applying techniques to detect and correct errors.
- There are various kinds of framing methods, such as character count, character stuffing, bit stuffing, and flag bytes.
- Character count framing uses a field in the header to indicate the number of characters in the frame. This method is simple but vulnerable to errors if the count is corrupted or the frame is lost.
- Character stuffing framing uses a special character, such as ESC, to mark the start and end of each frame. If the data contains the same character, it is preceded by another ESC to avoid confusion. This method is more robust but requires more overhead.
- Bit stuffing framing uses a special bit pattern, such as 01111110, to mark the start and end of each frame. If the data contains five consecutive 1s, a 0 is inserted after them to avoid confusion. This method is efficient and reliable but requires bit-level processing.
- Flag byte framing uses a special byte, such as 7E, to mark the start and end of each frame. If the data contains the same byte, it is replaced by a different byte, such as 7D, and the original byte is appended after it. This method is simple and flexible but requires byte-level processing.

Error Detection and Correction in link layer in Computer Networks

- Error detection and correction in link layer is the process of identifying and re-transmitting data frames that have been corrupted or lost during transmission .
- Error detection and correction techniques are essential for reliable and efficient data communication over noisy and unreliable channels.
- Some of the common types of errors that can occur in data transmission are:

- Single bit errors: A single bit in a data frame is changed from 0 to 1 or vice versa.
- Multiple bit errors: Two or more bits in a data frame are changed from 0 to 1 or vice versa.
- Burst errors: A sequence of consecutive bits in a data frame are changed from 0 to 1 or vice versa.
- Some of the common techniques for error detection and correction in link layer are:
 - Parity checks: A parity bit is added to the data frame to make the number of 1s in the frame even or odd. The receiver checks the parity bit and detects a single bit error if the parity does not match .
 - Checksums: A checksum is a value calculated from the data frame using a simple arithmetic operation such as addition or XOR. The sender appends the checksum to the frame and the receiver verifies it by recalculating the checksum from the received frame. A mismatch in the checksum indicates an error in the frame .
 - Cyclic redundancy check (CRC): A CRC is a value computed from the data frame using a polynomial division. The sender appends the CRC to the frame and the receiver verifies it by performing the same polynomial division on the received frame. A nonzero remainder indicates an error in the frame .
- Error detection techniques can only detect the presence of errors in the data frame, but not correct them. Error correction techniques can both detect and correct errors in the data frame.
- Some of the common techniques for error correction in link layer are:
 - Automatic repeat request (ARQ): ARQ is a protocol that uses acknowledgments and timeouts to ensure reliable data delivery. The sender re-transmits the data frame if it does not receive an acknowledgment from the receiver within a specified time or if it receives a negative acknowledgment indicating an error in the frame .
 - Forward error correction (FEC): FEC is a technique that adds redundant bits to the data frame to enable the receiver to correct errors without requesting re-transmission. The sender encodes the data frame using an error-correcting code such as Hamming code or Reed-Solomon code and the receiver decodes the received frame using the same code .

Flow control in link layer in Computer Networks

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It regulates the amount of data that a sender can send before receiving an acknowledgment from the receiver .
- It prevents the sender from overwhelming the receiver with too much data or frames.
- It can be implemented at the data link layer using two methods: stop-and-

wait and sliding window.

- Stop-and-wait is a simple method where the sender sends one frame at a time and waits for an acknowledgment before sending the next frame.
- Sliding window is a more efficient method where the sender can send multiple frames within a window size without waiting for acknowledgments.
- The window size is the number of frames that can be sent or received at a time.
- The window size can be fixed or variable depending on the protocol.
- Ethernet flow control is an example of flow control at the data link layer using pause frames.
- Pause frames are special frames that can be sent by the receiver to the sender to request a temporary halt of data transmission.

Elementary Data Link Protocols in link layer in Computer Networks

- Data link layer protocols are designed to perform the basic functions of the data link layer, such as framing, error control and flow control.
- Framing is the process of dividing the bit-streams from the physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control is the process of detecting and correcting errors that may occur during transmission or reception of data frames.
- Flow control is the process of regulating the rate of data transmission between the sender and the receiver to avoid congestion or buffer overflow.
- Elementary data link layer protocols are divided into three categories, as follows:
 - Protocol 1: Unrestricted simplex protocol
 - * This protocol allows the sender to transmit data frames continuously without any feedback from the receiver.
 - * The receiver does not send any acknowledgment or control frames to the sender.
 - * This protocol is suitable for simplex channels, where data transmission is only in one direction.
 - * This protocol does not provide any error control or flow control mechanisms.
 - Protocol 2: Simplex stop-and-wait protocol
 - * This protocol allows the sender to transmit one data frame at a time and wait for an acknowledgment from the receiver before sending the next frame.
 - * The receiver sends an acknowledgment frame to the sender after receiving and processing each data frame.
 - * This protocol is suitable for half-duplex channels, where data transmission can occur in both directions, but not simultaneously.
 - * This protocol provides error control by using sequence numbers

- and retransmission of lost or damaged frames.
 - * This protocol provides flow control by using a stop-and-wait mechanism, where the sender cannot send more than one frame at a time.
- Protocol 3: Simplex protocol for noisy channels
 - * This protocol is an improvement of protocol 2, which can handle noisy channels, where data frames or acknowledgment frames may be lost or corrupted.
 - * This protocol uses timers and timeouts to detect lost or delayed frames or acknowledgments.
 - * This protocol uses negative acknowledgments (NAKs) to indicate corrupted frames or acknowledgments.
 - * This protocol uses piggybacking to combine data frames and acknowledgment frames into one frame, to reduce the overhead and improve the efficiency.
 - * This protocol is suitable for half-duplex or full-duplex channels, where data transmission can occur in both directions, either alternately or simultaneously.
 - * This protocol provides error control and flow control by using sequence numbers, retransmission, timers, timeouts, NAKs and piggybacking.
- Some examples of data link layer protocols are:
 - Synchronous Data Link Control (SDLC): A bit-oriented protocol that uses flags, bit stuffing and cyclic redundancy check (CRC) for framing and error control.
 - High-Level Data Link Control (HDLC): A bit-oriented protocol that is based on SDLC, but uses different frame types and control fields for different functions.
 - Serial Line Internet Protocol (SLIP): A character-oriented protocol that uses end-of-frame markers and checksum for framing and error control.
 - Point-to-Point Protocol (PPP): A byte-oriented protocol that uses flags, byte stuffing and CRC for framing and error control, and also supports multiple network layer protocols.
 - Link Control Protocol (LCP): A protocol that is used to establish, configure, test and terminate the data link connection in PPP.
 - Link Access Procedure (LAP): A protocol that is used to provide error control and flow control for data link layer protocols, such as LAPB and LAPD.
 - Network Control Protocol (NCP): A protocol that is used to negotiate and configure the network layer protocols in PPP, such as IP, IPX and AppleTalk.

Sliding Window protocols in link layer in Computer Networks

- The sliding window protocol is a data link layer protocol that is useful in

the sequential and reliable delivery of the data frames .

- Using the sliding window protocol, the sender can send multiple frames at a time before receiving an acknowledgment from the receiver .
- The sliding window is also used in Transmission Control Protocol (TCP), which operates at the transport layer .
- The sliding window protocol manages the flow of data between two network nodes to ensure that the receiver can handle the incoming data and that the sender does not overwhelm the network .
- The sliding window protocol uses two types of windows: a send window and a receive window .
- The send window is the set of frames that the sender can transmit without waiting for an acknowledgment .
- The receive window is the set of frames that the receiver can accept without sending an acknowledgment .
- The size of the send window and the receive window can vary depending on the network conditions and the protocol parameters .
- The sliding window protocol can be classified into two types: stop-and-wait and go-back-N .
- In stop-and-wait, the sender sends one frame at a time and waits for an acknowledgment before sending the next frame .
- In go-back-N, the sender can send up to N frames at a time without waiting for an acknowledgment, where N is the size of the send window .
- The receiver sends an acknowledgment for the last frame it received in order, and the sender retransmits all the frames from that point onwards in case of a lost or corrupted frame .
- The sliding window protocol improves the efficiency and reliability of data transmission over a network .

Medium Access Control and Local Area Networks

- Medium Access Control (MAC) is a sublayer of the data link layer that regulates the access of multiple devices to a shared medium, such as a wireless channel or a wired network.
- MAC protocols are designed to avoid or resolve collisions, which occur when two or more devices transmit data at the same time on the same medium.
- There are two main approaches to MAC in Local Area Networks (LANs): contention and token-passing.
 - Contention is a first-come, first-serve approach, where devices compete for the medium and back off when they detect a collision. An example of a contention-based MAC protocol is Carrier Sense Multiple Access with Collision Detection (CSMA/CD), which is used in Ethernet networks.
 - Token-passing is a cooperative approach, where devices take turns to access the medium by passing a token among themselves. An example of a token-passing MAC protocol is Token Ring, which was

used in some early LANs.

- MAC protocols can also be classified as centralized or distributed, depending on whether there is a central controller or not that coordinates the access to the medium.
 - Centralized MAC protocols rely on a master device or an access point that grants or denies the requests of other devices to access the medium. An example of a centralized MAC protocol is Wireless Fidelity (Wi-Fi), which uses a Distributed Coordination Function (DCF) based on CSMA/CA (Collision Avoidance) and an optional Point Coordination Function (PCF) based on polling.
 - Distributed MAC protocols allow devices to access the medium without any coordination from a central controller. An example of a distributed MAC protocol is Bluetooth, which uses a Frequency Hopping Spread Spectrum (FHSS) technique and a Time Division Multiple Access (TDMA) scheme.
- MAC protocols are designed to meet the specific requirements and characteristics of different LANs, such as the physical layer, the topology, the traffic patterns, the quality of service, the security, and the scalability.

Channel allocation in medium access control

- Channel allocation is the process of assigning different frequency bands or time slots to different users or devices that want to communicate over a shared medium.
- Medium access control (MAC) is the protocol or mechanism that coordinates the access to the channel, while ensuring good throughput, fairness, low latency and low energy consumption.
- Channel allocation and medium access control are closely related, as the MAC protocol may depend on the channel allocation scheme, and vice versa.
- There are different types of channel allocation schemes, such as:
 - Fixed channel allocation: Each user or device is assigned a fixed channel or set of channels for the duration of the communication. This scheme is simple and avoids collisions, but may waste bandwidth if some channels are underutilized or some users have high demand.
 - Dynamic channel allocation: The channel or set of channels is assigned to each user or device on demand, based on some criteria such as traffic load, priority, channel availability, etc. This scheme is more flexible and efficient, but may incur higher overhead and complexity.
 - Hybrid channel allocation: A combination of fixed and dynamic channel allocation, where some channels are reserved for certain users or devices, and some channels are shared among others. This scheme can balance the trade-off between simplicity and flexibility, but may require coordination and synchronization among users or devices.
- There are different types of MAC protocols, such as:
 - Contention-based MAC: Each user or device competes for the channel

access, using some random or deterministic algorithm to avoid or resolve collisions. Examples of contention-based MAC protocols are ALOHA, CSMA, CSMA/CD, CSMA/CA, etc.

- Reservation-based MAC: Each user or device reserves the channel access in advance, using some signaling or negotiation mechanism. Examples of reservation-based MAC protocols are TDMA, FDMA, CDMA, OFDMA, etc.
- Hybrid MAC: A combination of contention-based and reservation-based MAC, where some parts of the channel are accessed by contention, and some parts are accessed by reservation. Examples of hybrid MAC protocols are IEEE 802.11, IEEE 802.15.4, etc.

Multiple access protocols in medium access control

- Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC sublayer) of the Open Systems Interconnection (OSI) model .
- These protocols allow a number of nodes or users to access a shared network channel, such as a wireless channel or a cable network .
- The main objectives of multiple access protocols are to avoid or resolve collisions, to allocate the channel efficiently, and to ensure fairness among the nodes or users .
- There are different types of multiple access protocols, such as random access protocols, controlled access protocols, and channelization protocols .
- Random access protocols allow any node or user to transmit data at any time, depending on the state of the channel (idle or busy). Examples of random access protocols are ALOHA, Carrier Sense Multiple Access (CSMA), CSMA with Collision Avoidance (CSMA/CA), and CSMA with Collision Detection (CSMA/CD) .
- Controlled access protocols require a node or user to obtain permission before transmitting data. Examples of controlled access protocols are Reservation, Polling, and Token Passing .
- Channelization protocols divide the channel into smaller units, such as time slots, frequency bands, or codes, and assign them to different nodes or users. Examples of channelization protocols are Time Division Multiple Access (TDMA), Frequency Division Multiple Access (FDMA), and Code Division Multiple Access (CDMA) .
- Some multiple access protocols are specific to certain network technologies, such as the Packet Data Convergence Protocol (PDCP), the Radio Link Control (RLC) protocol, and the MAC protocol for UMTS and LTE.

LAN standards in local area network

- A local area network (LAN) is a data communication network connecting various terminals or computers within a building or limited geographical

area.

- The connection among the devices could be wired or wireless.
- LAN standards are the specifications and protocols that define the operation and performance of a LAN.
- The most widely used family of LAN standards is IEEE 802, which covers different types of LAN technologies, such as Ethernet, Token Ring, Wireless LAN, etc .
- Each LAN standard has a number and a name, such as IEEE 802.3 for Ethernet, IEEE 802.11 for Wireless LAN, IEEE 802.15 for Wireless PAN, etc .
- Each LAN standard defines the physical layer and the data link layer of the network, as well as the media access control (MAC) method, the frame format, the addressing scheme, the error detection and correction, and the network management .
- LAN standards are important for ensuring interoperability, compatibility, and reliability of LAN devices and networks .

Link layer switches & bridges in local area network

- Link layer switches and bridges are network devices that operate at the data link layer (layer 2) of the OSI model and connect multiple LANs (local area networks) together to form a larger LAN.
- Link layer switches and bridges can forward Ethernet frames from one device to another device based on their MAC addresses, which are unique identifiers for each device on a LAN.
- Link layer switches and bridges can also filter and block frames that are not destined for the other LANs, thus reducing network congestion and improving network performance.
- Link layer switches and bridges can interconnect LANs that use different data link layer technologies, such as Ethernet, FDDI, or Token Ring, by converting the frame formats and adjusting the maximum frame size if needed.
- Link layer switches and bridges can learn the MAC addresses of the devices on each LAN by observing the source addresses of the incoming frames and storing them in a table. This table is used to determine the destination port for each frame.
- Link layer switches and bridges can operate in different modes, such as store-and-forward, cut-through, or fragment-free, depending on how they process the frames. Store-and-forward mode involves storing the entire frame in memory and checking it for errors before forwarding it. Cut-through mode involves forwarding the frame as soon as the destination address is read, without checking for errors. Fragment-free mode involves storing the first 64 bytes of the frame and checking for errors before forwarding it.

Learning bridge algorithms in local area network

- A bridge is a device that connects two or more local area networks (LANs) and forwards frames based on their destination MAC addresses.
- A bridge algorithm is a set of rules or procedures that a bridge uses to decide which frames to forward, which frames to filter, and which ports to use.
- There are different types of bridge algorithms, such as transparent bridging, source routing bridging, and spanning tree protocol.
- Transparent bridging is the most common type of bridge algorithm. It uses a self-learning technique to build and maintain a forwarding table that maps MAC addresses to ports.
- Source routing bridging is used in token ring networks. It requires the source station to specify the path of the frame in the header, using a route discovery process.
- Spanning tree protocol is a bridge algorithm that prevents loops in a network with redundant links. It creates a logical tree topology by blocking some ports and allowing only one path between any two LANs.

Spanning Tree Algorithms in Local Area Network

- Spanning tree algorithms are used to prevent loops in a network topology that uses bridges or switches to connect multiple segments of a local area network (LAN).
- Loops can cause problems such as broadcast storms, multiple frame transmission, and MAC address table instability.
- Spanning tree algorithms work by creating a logical tree structure of the network, where only one path exists between any two nodes. The tree is rooted at a designated bridge or switch, called the root bridge.
- The algorithm blocks or disables the links that are not part of the tree, thus breaking the loops. The blocked links can be used as backup in case of link failure.
- The most common spanning tree algorithm is the Spanning Tree Protocol (STP), which is standardized by IEEE 802.1D. STP uses a set of rules to determine the root bridge, the role of each bridge or switch, and the state of each port.
- STP uses the following parameters to select the root bridge and the best path to the root:
 - Bridge ID: a combination of bridge priority and MAC address. The bridge with the lowest bridge ID becomes the root bridge.
 - Path cost: a metric that reflects the bandwidth of a link. The path with the lowest total cost to the root is preferred.
 - Port ID: a combination of port priority and port number. The port with the lowest port ID is preferred in case of a tie.
- STP assigns one of the following roles to each port of a bridge or switch:
 - Root port: the port that has the best path to the root bridge. There is only one root port per bridge or switch, except the root bridge itself.

- Designated port: the port that has the best path to the root bridge among all the ports on the same segment. There is only one designated port per segment, except the root bridge itself.
- Alternate port: a backup port that provides an alternative path to the root bridge, in case the root port fails. An alternate port is blocked by default, but can become a root port if needed.
- Backup port: a backup port that provides an alternative path to the designated port on the same segment, in case the designated port fails. A backup port is blocked by default, but can become a designated port if needed.
- Disabled port: a port that is manually shut down or administratively disabled. A disabled port does not participate in STP.
- STP assigns one of the following states to each port of a bridge or switch, based on its role and the network topology:
 - Blocking: the port is not forwarding frames, and does not learn MAC addresses. The port only receives and processes STP messages, called Bridge Protocol Data Units (BPDUs). This is the initial state of all ports, except the root bridge ports.
 - Listening: the port is not forwarding frames, and does not learn MAC addresses. The port receives and processes BPDUs, and also sends BPDUs to announce its role and state. The port transitions to this state from blocking, when it becomes a root port or a designated port.
 - Learning: the port is not forwarding frames, but learns MAC addresses from the frames it receives. The port receives and processes BPDUs, and also sends BPDUs to announce its role and state. The port transitions to this state from listening, after a timer called forward delay expires.
 - Forwarding: the port is forwarding frames, and learns MAC addresses from the frames it receives. The port receives and processes BPDUs, and also sends BPDUs to announce its role and state. The port transitions to this state from learning, after another timer called forward delay expires. This is the normal state of a root port or a designated port.
 - Disabled: the port is not forwarding frames, and does not learn MAC addresses. The port does not receive or process BPDUs, and does not send BPDUs. The port transitions to this state from any other state, when it is manually shut down or administratively disabled.
- STP uses timers to control the frequency and duration of BPDUs, and to prevent the network from changing too frequently or too slowly. The timers are:
 - Hello time: the interval between two consecutive BPDUs sent by a bridge or switch. The default value is 2 seconds.
 - Max age: the maximum time that a bridge or switch can store a BPDU received from another bridge or switch, before discarding it. The default value is 20 seconds.

- Forward delay: the time that a

Unit 3 - Network Layer in Computer Networks

The network layer is responsible for routing packets from the source to the destination across multiple hops and networks. It also provides services such as addressing, congestion control, error detection and correction, fragmentation and reassembly, and quality of service.

Some of the main topics covered in this unit are:

- Network layer functions and design issues: This topic explains the role of the network layer in the Internet architecture, the principles of network layer design, and the challenges faced by network layer protocols.
- Network layer protocols: This topic introduces the main network layer protocols used in the Internet, such as IP, ICMP, ARP, DHCP, and NAT. It also describes the format, functions, and operations of these protocols.
- Routing algorithms: This topic covers the basic concepts and techniques of routing, such as routing tables, forwarding, and routing metrics. It also discusses the types and characteristics of routing algorithms, such as static, dynamic, centralized, distributed, adaptive, and non-adaptive.
- Routing protocols: This topic explains the operation and features of various routing protocols used in the Internet, such as RIP, OSPF, BGP, and multicast routing protocols. It also compares and contrasts the advantages and disadvantages of these protocols.
- Congestion control and quality of service: This topic addresses the problem of congestion in networks, and the methods and mechanisms to prevent, detect, and control congestion. It also introduces the concept and requirements of quality of service, and the techniques and protocols to provide different levels of service to different applications and users.

Point-to-point networks in network layer

- A point-to-point network is a network topology that connects two nodes directly using a single link.
- A point-to-point network can be used to connect two routers, two hosts, or a router and a host without any intermediate devices or networks.
- A point-to-point network can provide high bandwidth, low latency, and reliable communication between the two nodes.
- A point-to-point network can also support authentication, encryption, and compression of the data transmitted over the link.
- A common protocol used for point-to-point networks is the Point-to-Point Protocol (PPP), which operates at the data link layer (layer 2) of the OSI model.
- PPP can encapsulate different types of network layer (layer 3) packets, such as IP, IPX, or AppleTalk, into frames for transmission over the point-to-point link.

- PPP also has a Link Control Protocol (LCP) that establishes, configures, and tests the data link connection, and a Network Control Protocol (NCP) that negotiates the network layer parameters, such as IP address, for each encapsulated protocol.
- PPP can also use other protocols, such as Point-to-Point Tunneling Protocol (PPTP), to create virtual point-to-point links over other networks, such as the Internet.

Logical addressing in network layer

- Logical addressing is a way of identifying devices on a network using addresses that are assigned by a network layer protocol, such as IP or IPX .
- Logical addresses are also called network addresses or layer 3 addresses.
- Logical addresses are independent of the physical addresses, such as MAC addresses, that are used by the data link layer or layer 2 .
- Logical addresses allow devices to communicate across different physical networks, as the network layer can translate logical addresses to physical addresses using a process called routing .
- Logical addresses are usually hierarchical, meaning they have two parts: a network part and a host part. The network part identifies the network or subnet that the device belongs to, and the host part identifies the specific device within that network or subnet .
- Logical addresses can have different formats and lengths depending on the network layer protocol. For example, IPv4 addresses are 32 bits long and are usually written in dotted decimal notation, such as 192.168.1.1. IPv6 addresses are 128 bits long and are usually written in hexadecimal notation, such as 2001:db8::1 .
- Logical addresses are assigned to devices either statically or dynamically. Static addressing means the address is manually configured by an administrator and does not change. Dynamic addressing means the address is automatically obtained from a server or a protocol and can change over time .

Basic internetworking in network layer

- Internetworking is the process of connecting different networks using routers and other devices that operate at the network layer of the OSI model .
- An internetwork is a collection of networks that can communicate with each other using a common protocol, such as the Internet Protocol (IP).
- Internetworking enables data exchange and resource sharing across heterogeneous networks, such as Ethernet, wireless, and point-to-point links.
- Internetworking requires addressing and routing mechanisms to identify and deliver packets to the correct destination network and host.
- There are three types of internetwork addresses that are commonly used:

data-link layer addresses, media access control (MAC) addresses, and network-layer addresses.

- Data-link layer addresses are used to identify each physical network connection of a network device, such as a network interface card (NIC).
- MAC addresses are a type of data-link layer addresses that are assigned by the manufacturer of the NIC and are usually fixed and unique.
- Network-layer addresses are logical addresses that are assigned by the network administrator or the network service provider and can be changed or reassigned.
- IP addresses are a type of network-layer addresses that are used to identify each host and network in an internetwork.
- Routers are the devices that interconnect different networks and perform the internetworking functions of addressing and routing.
- Routers use routing tables and routing protocols to determine the best path for each packet to reach its destination.
- The Internet is the most notable example of an internetwork that connects millions of networks and hosts using the TCP/IP protocol suite .
- The Internet is composed of four layers: the network access layer, the internetwork layer, the host-to-host transport layer, and the application layer.
- The network access layer defines the physical and data-link layer specifications for different types of networks, such as Ethernet, wireless, and point-to-point links.
- The internetwork layer defines the IP protocol and its related functions, such as addressing, fragmentation, and error handling.
- The host-to-host transport layer defines the TCP and UDP protocols and their related functions, such as reliability, flow control, and port numbers.
- The application layer defines the protocols and services that are used by applications to communicate with each other, such as HTTP, FTP, and SMTP.

IP

IP can have different meanings depending on the context. Here are some possible definitions of IP:

- **Intellectual property:** This is a legal term that refers to creations of the mind, such as inventions, literary and artistic works, designs, symbols, names and images used in commerce. Intellectual property is protected by laws such as patents, copyrights and trademarks, which enable people to earn recognition or financial benefit from what they invent or create.
- **Internet protocol:** This is a technical term that refers to a set of rules that enable the modern internet. It specifies the format for addresses and units of transmitted data. Every computer, modem, server and smart-phone that connects to the internet has a unique IP address, which helps users to find it. Internet protocol is often used with another protocol called

transmission control protocol (TCP), which ensures the reliable delivery of data over the internet.

- **Innings pitched:** This is a sports term that refers to a statistic in baseball and softball. It measures the number of innings a pitcher has completed, from the start of a game until he or she is replaced by another pitcher. One inning is equal to three outs, so if a pitcher retires three batters in a row, he or she has pitched one inning. Innings pitched is used to calculate other statistics such as earned run average (ERA) and strikeouts per nine innings ($K/9$).

CIDR

- CIDR stands for Classless Inter-Domain Routing, a method for allocating IP addresses and for IP routing .
- CIDR replaces the previous classful network addressing architecture on the Internet, which was based on fixed classes A, B and C .
- CIDR allows blocks of IP addresses to be grouped into single routing table entries, which reduces the size and complexity of routing tables and improves the efficiency of address distribution .
- CIDR notation is a compact representation of an IP address and its associated routing prefix. It consists of an IP address followed by a slash (/) and a number, called the prefix length, which indicates the number of bits in the network portion of the address .
- For example, 192.168.1.0/24 is a CIDR notation for a block of 256 IP addresses, from 192.168.1.0 to 192.168.1.255, where the first 24 bits are the network prefix and the last 8 bits are the host identifier.
- CIDR notation can also be used to specify a single IP address, by using a prefix length of 32. For example, 192.168.1.1/32 is equivalent to 192.168.1.1.
- CIDR notation is widely used in network configuration, routing protocols, firewall rules, and access control lists .

ARP

- ARP stands for Address Resolution Protocol .
- It is a communication protocol used for discovering the link layer address, such as a MAC address, associated with a given internet layer address, typically an IPv4 address .
- It is a critical function in the Internet protocol suite, as it enables devices to communicate within a local area network (LAN) .
- ARP was defined in 1982 by RFC 826, which is Internet Standard STD 37.
- ARP works by sending broadcast messages to all devices in the LAN, asking for the MAC address of the device that has a specific IP address.
- The device that has the requested IP address replies with its MAC address, and the sender updates its ARP cache with the mapping of the IP address

and the MAC address.

- ARP cache is a table that stores the mappings of IP addresses and MAC addresses for a certain period of time.
- ARP can be divided into two types: static ARP and dynamic ARP.
- Static ARP is when the mappings of IP addresses and MAC addresses are manually entered and do not change.
- Dynamic ARP is when the mappings of IP addresses and MAC addresses are automatically learned and updated by ARP messages.
- ARP can also be classified into four types based on the types of addresses involved: ARP, RARP, InARP and Proxy ARP.
- ARP is when the sender knows the IP address of the destination and wants to find its MAC address.
- RARP is when the sender knows the MAC address of the destination and wants to find its IP address.
- InARP is when the sender knows the MAC address of a device on another network and wants to find its IP address.
- Proxy ARP is when a device acts as an intermediary and responds to ARP requests on behalf of another device.

RARP

- RARP stands for Reverse Address Resolution Protocol .
- It is a protocol based on computer networking that is used by a client computer to request its IP address from a gateway server's Address Resolution Protocol (ARP) table or cache.
- It works on the Network Access Layer (the lowest layer) of the TCP/IP protocol stack.
- It operates by sending the device's physical address (MAC address) to a specialized RARP server that is on the same local area network (LAN) and is actively listening for RARP requests.
- The RARP server then looks up the MAC address in its ARP table or cache and returns the corresponding IP address to the client computer.
- RARP is useful for devices that do not have a permanent IP address, such as diskless workstations or network printers .
- RARP is an obsolete protocol that has been replaced by other methods of obtaining IP addresses, such as BOOTP and DHCP .

DHCP

- DHCP stands for Dynamic Host Configuration Protocol. It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices connected to a network using a client-server architecture .
- DHCP operates on four basic steps: discover, offer, request, and acknowledge. A client device sends a broadcast message (discover) to the network, requesting an IP address. A DHCP server responds with an offer message,

containing an available IP address and other configuration information. The client device accepts the offer and sends a request message to the server. The server acknowledges the request and assigns the IP address to the client device .

- DHCP has several advantages over manual IP configuration, such as reducing network administration, avoiding IP conflicts, improving network security, and supporting network mobility .
- DHCP can also provide other information to the client devices, such as the subnet mask, default gateway, domain name, DNS servers, and time servers .
- DHCP is defined by RFCs 2131 and 2132, and is an Internet Engineering Task Force (IETF) standard based on Bootstrap Protocol (BOOTP) .

ICMP

- ICMP stands for Internet Control Message Protocol .
- It is a network layer protocol that is used by network devices to communicate problems with data transmission .
- It is mainly used to determine whether data is reaching its intended destination in a timely manner .
- It also performs network diagnostics and reports errors in IP operations .
- ICMP messages are encapsulated in IP datagrams and have a specific format that includes a type, a code, a checksum, and optional data fields.
- Some common ICMP message types are:
 - Echo request and echo reply: used to test the reachability and round-trip time of a destination .
 - Destination unreachable: used to indicate that a destination or a service is unreachable for some reason .
 - Time exceeded: used to indicate that a datagram has expired in transit or in a reassembly queue .
 - Parameter problem: used to indicate that a datagram has an invalid or missing option or header field.
 - Redirect: used to inform a source host of a better route to a destination.
 - Router advertisement and router solicitation: used to discover and advertise routers on a network.
- ICMP is part of the Internet protocol suite as defined in RFC 792.

Routing in network layer

- Routing is the process of finding a path from a source to a destination in a network of nodes, such as routers, switches, or hosts.
- Routing is performed by network layer protocols, such as IP, that operate on logical addresses, such as IP addresses, rather than physical addresses, such as MAC addresses.
- Routing can be classified into two types: static routing and dynamic routing.

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- Static routing is when the routes are manually configured by the network administrator and do not change unless updated manually. Static routing is simple, secure, and predictable, but it is not scalable, adaptable, or efficient.
- Dynamic routing is when the routes are automatically learned and updated by the network devices using routing protocols, such as RIP, OSPF, or BGP. Dynamic routing is scalable, adaptable, and efficient, but it is more complex, less secure, and less predictable.
- Routing protocols can be further classified into two types: distance vector protocols and link state protocols.
- Distance vector protocols are based on the principle of exchanging distance information, such as hop count or cost, with neighboring nodes. Each node maintains a routing table that contains the best known distance and next hop for each destination. Examples of distance vector protocols are RIP and BGP.
- Link state protocols are based on the principle of exchanging topology information, such as link status and cost, with all nodes in the network. Each node maintains a link state database that contains the complete network topology. Each node then runs a shortest path algorithm, such as Dijkstra's algorithm, to compute the best path for each destination. Examples of link state protocols are OSPF and IS-IS.

Forwarding and Delivery in Network Layer

- The network layer supervises the handling of packets by the underlying physical networks. We call this handling as the delivery of packets to the destination .
- The delivery of a packet is called **direct** if the deliverer (host or router) and the destination are on the same network; the delivery of a packet is called **indirect** if the deliverer (host or router) and the destination are on different networks.
- Forwarding means placing the packet in its route to the destination and it requires a routing table . A routing table is a data structure that stores the information about the best routes to reach each destination network.
- Forwarding refers to the router-local action of transferring a packet from an input link interface to the appropriate output link interface. Routing refers to the network-wide process that determines the end-to-end paths that packets take from source to destination.
- Address aggregation is a technique that reduces the size of routing tables by grouping several networks into a single entry. For example, if a router has four networks with the same prefix, it can aggregate them into one entry with a shorter prefix length.
- Some tools or utilities that can help with packet delivery and routing are :
 - **ping**: a command that sends an echo request packet to a destination

and waits for an echo reply packet. It can measure the round-trip time and check the connectivity between two hosts or routers.

- **tracert**: a command that traces the route taken by a packet from the source to the destination. It can show the number of hops, the IP addresses of intermediate routers, and the delay of each hop.
- **route**: a command that displays or modifies the routing table of a host or router. It can show the destination network, the gateway, the interface, and the metric for each entry.

Static and dynamic routing in computer networks

- Static routing and dynamic routing are two methods used to determine how to send a packet toward its destination.
- Static routes are configured in advance of any network communication by the network administrator .
- Dynamic routes are learned by routers through exchanging information with other routers using complex routing algorithms .

Advantages and disadvantages of static routing

- Static routing provides more security as the routes are not advertised to other routers.
- Static routing is more efficient as it does not consume bandwidth or CPU resources for exchanging routing information.
- Static routing is easier to implement and troubleshoot in small networks .
- Static routing is less adaptable to network changes or failures as the routes have to be manually reconfigured .
- Static routing is not scalable for large networks as it requires a lot of manual work and maintenance .

Advantages and disadvantages of dynamic routing

- Dynamic routing provides more flexibility and reliability as it can adjust routes automatically based on network conditions or topology changes .
- Dynamic routing is more scalable for large networks as it reduces the administrative burden and complexity of routing configuration .
- Dynamic routing provides less security as the routes are advertised to other routers and may be vulnerable to attacks or misconfigurations.
- Dynamic routing is less efficient as it consumes bandwidth and CPU resources for exchanging routing information and running routing algorithms.
- Dynamic routing is harder to implement and troubleshoot in complex networks as it requires a good understanding of the routing protocols and their parameters.

Types of static routes

- Standard static route: a static route that specifies the next-hop address or interface to reach a specific destination network.
- Default static route: a static route that matches all packets that do not have a more specific route in the routing table and sends them to a default gateway.
- Summary static route: a static route that aggregates multiple subnets into a single route and reduces the size of the routing table.
- Floating static route: a static route that has a higher administrative distance than a dynamic route and acts as a backup route in case the dynamic route fails.

Routing algorithms and protocols in computer networks

- Routing is the process of finding and selecting the best path for data transmission from source to destination in a network.
- Routing algorithms are the software programs that implement the logic of routing, i.e., they decide the optimal path for each packet based on some criteria or metrics.
- Routing protocols are the set of rules and procedures that routers use to communicate with each other and exchange routing information.
- Routing algorithms and protocols can be classified into three major categories :
 - Interior gateway protocols (IGPs): These are the routing protocols used within a single autonomous system (AS), which is a group of routers under the same administrative control. IGPs can be further divided into two types:
 - * Link-state routing protocols: These protocols maintain a complete map of the network topology and calculate the shortest path to each destination using an algorithm like Dijkstra's or Bellman-Ford. Examples of link-state routing protocols are Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS).
 - * Distance-vector routing protocols: These protocols only keep track of the distance and direction to each destination and exchange this information with their neighbors periodically. Examples of distance-vector routing protocols are Routing Information Protocol (RIP), Enhanced Interior Gateway Routing Protocol (EIGRP), and Border Gateway Protocol (BGP).
 - Exterior gateway protocols (EGPs): These are the routing protocols used between different autonomous systems. They are responsible for inter-domain routing and policy enforcement. The most widely used EGP is BGP, which is also a distance-vector protocol.
 - Hybrid routing protocols: These are the routing protocols that combine the features of both link-state and distance-vector protocols. They can adapt to different network scenarios and provide better performance and scalability. Examples of hybrid routing protocols

are Enhanced Interior Gateway Routing Protocol (EIGRP) and Zone Routing Protocol (ZRP).

- Routing algorithms and protocols use various metrics to determine the best path for each packet, such as distance, hop count, bandwidth, delay, load, reliability, cost, etc. Different metrics may suit different network objectives and constraints.
- Routing algorithms and protocols also have to deal with various challenges and trade-offs, such as convergence, loop prevention, scalability, robustness, security, etc. Different algorithms and protocols may have different strengths and weaknesses in handling these issues .

Congestion control algorithms in computer networks

- Congestion control algorithms are mechanisms that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- Congestive collapse is a situation where the network performance degrades drastically due to excessive traffic and congestion.
- Congestion control algorithms can be broadly classified into two categories: open loop and closed loop.
- Open loop congestion control policies are applied to prevent congestion before it happens. They involve designing the network and the protocols in such a way that congestion is unlikely to occur.
- Closed loop congestion control policies are applied to detect and mitigate congestion after it happens. They involve monitoring the network state and adjusting the transmission rate or window size of the senders based on feedback signals such as packet losses and delays.
- Some examples of open loop congestion control techniques are admission control, traffic shaping, and resource reservation.
- Some examples of closed loop congestion control techniques are congestion avoidance, congestion recovery, and congestion pricing.
- Congestion avoidance algorithms are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. They use a combination of additive increase/multiplicative decrease (AIMD), slow start, and congestion window (CWND) schemes to adjust the transmission rate of the senders based on the network conditions.
- Congestion recovery algorithms are implemented at the TCP layer as the mechanism to recover from packet losses due to congestion. They use a combination of fast retransmit, fast recovery, and selective acknowledgment (SACK) schemes to retransmit the lost packets and resume the transmission rate of the senders based on the network conditions.
- Congestion pricing algorithms are implemented at the network layer as the mechanism to allocate the network resources based on the demand and supply of the users. They use a combination of pricing, auction, and incentive schemes to charge the users for the network usage and encourage them to reduce their demand during congestion.

- Some examples of congestion avoidance algorithms are TCP Tahoe, TCP Reno, TCP New Reno, TCP Vegas, TCP BIC, and TCP CUBIC.
- Some examples of congestion recovery algorithms are TCP SACK, TCP FACK, and TCP D-SACK.
- Some examples of congestion pricing algorithms are Smart Market, Progressive Second Price, and Proportional Fairness.

IPv6 in China

- IPv6 is the latest version of the Internet Protocol, which provides virtually unlimited IP addresses for electronic devices to communicate online.
- China is one of the leading countries in IPv6 deployment and application, with the goal of running a single-stack IPv6 network by 2030.
- China has over 900 million mobile subscribers and over 670 million 4G/LTE subscribers, which require a large number of IP addresses.
- China has implemented a series of policies and measures to promote IPv6, such as issuing IPv6 licenses, allocating IPv6 address resources, building IPv6 backbone networks, upgrading IPv6 applications and services, and enhancing IPv6 security and testing .
- China has achieved remarkable progress in IPv6 adoption, with more than 700 million active IPv6 users and 200 million IoT devices using IPv6 as of 2023 .
- China also aims to shape the IPv6 standards and contribute to the global IPv6 development, by participating in international organizations, forums, and events, and by conducting research and innovation on IPv6 technologies.

Unit 4 - Transport Layer in Computer Networks

- The transport layer is a conceptual division of methods in the layered architecture of protocols in the network stack in the Internet protocol suite and the OSI model.
- The protocols of this layer provide end-to-end communication services for applications, such as reliable data transfer, flow control, congestion control, multiplexing, and error detection .
- The transport layer takes data from the upper layer (i.e. application layer) and then breaks it into smaller size segments, numbers each byte, and hands over to the lower layer (network layer) for delivery.
- The transport layer also reassembles the segments at the destination and delivers the complete message to the upper layer.
- The transport layer provides the user address which is specified as a station or port. The port variable represents a specific process within a host that is running an application program.
- The transport layer protocols need to know the network layer address of the destination host to deliver the segments. This is done by using a name resolution service, such as DNS, to map the user-friendly name to

the network layer address.

- The transport layer can be either connection-oriented or connectionless, depending on the protocol used. The connection-oriented protocols establish a logical connection between the source and destination before transferring data, such as TCP. The connectionless protocols do not require a connection establishment and send data as independent datagrams, such as UDP.
- The transport layer can also provide different levels of service quality, such as throughput, delay, jitter, and reliability, depending on the application requirements and the network conditions.

Process-to-process delivery in transport layer

- The transport layer is the fourth layer of the OSI model, which provides services to the application layer.
- The transport layer is responsible for process-to-process delivery, which means the delivery of a packet, part of a message, from one process to another.
- A process is an entity of the application layer that uses the services of the transport layer. Two processes can communicate in a client/server relationship.
- The transport layer uses two methods to identify the processes: port numbers and service access points (SAPs).
- A port number is a 16-bit number that uniquely identifies a process on a host. There are two types of port numbers: well-known ports and registered ports.
- A well-known port is a port number assigned to a common service, such as HTTP (80), FTP (21), or Telnet (23). These port numbers are standardized by the Internet Assigned Numbers Authority (IANA).
- A registered port is a port number assigned to a specific service or application by the IANA upon request. These port numbers range from 1024 to 49151.
- A service access point (SAP) is a logical address that identifies a process on a host. A SAP consists of a network address and a port number. For example, a SAP for a web server on a host with IP address 192.168.1.10 and port number 80 is 192.168.1.10:80.
- The transport layer uses two protocols to provide process-to-process delivery: TCP and UDP.
- TCP (Transmission Control Protocol) is a connection-oriented, reliable, and full-duplex protocol that provides error control, flow control, and congestion control. TCP establishes a logical connection between two processes before exchanging data, and ensures that the data is delivered in the same order and without errors.
- UDP (User Datagram Protocol) is a connectionless, unreliable, and simplex protocol that provides no error control, flow control, or congestion control. UDP does not establish a connection between two processes, and

does not guarantee the delivery, order, or integrity of the data. UDP is used for applications that require speed and efficiency, such as streaming media or online gaming.

Transport layer protocols

- Transport layer protocols are responsible for providing end-to-end communication services for applications over a network.
- Transport layer protocols lie between the application layer and the network layer in the protocol stack.
- Transport layer protocols use port numbers to identify different applications or processes that communicate with each other.
- Transport layer protocols can be classified into two types: connection-oriented and connectionless.
- Connection-oriented protocols establish a logical connection between the sender and the receiver before exchanging data, and ensure reliable and ordered delivery of data. An example of a connection-oriented protocol is the Transmission Control Protocol (TCP).
- Connectionless protocols do not require a prior connection between the sender and the receiver, and do not guarantee reliable and ordered delivery of data. An example of a connectionless protocol is the User Datagram Protocol (UDP).
- Some additional transport layer protocols that have been defined and implemented are the Datagram Congestion Control Protocol (DCCP) and the Stream Control Transmission Protocol (SCTP), which offer different trade-offs between reliability, congestion control, and multiplexing.

UDP Transport layer protocol

- UDP stands for User Datagram Protocol. It is a transport layer protocol that is part of the Internet Protocol suite, also known as UDP/IP suite .
- UDP is a simple, unreliable and connectionless protocol. It does not establish a connection before sending data, nor does it guarantee the delivery, order or integrity of the data .
- UDP has a 16-byte header that contains the source and destination port numbers, the length of the data and a checksum for error detection . The header is shown below:

Source Port	Destination Port
16 bits	16 bits

Length	Checksum
16 bits	16 bits

- UDP is suitable for applications that require fast, low-overhead and real-time data transmission, such as voice and video streaming, online gaming, DNS queries and SNMP .
- UDP is also used by some transport layer protocols that provide additional features, such as Datagram Congestion Control Protocol (DCCP) and Stream Control Transmission Protocol (SCTP).

TCP Transport layer protocol

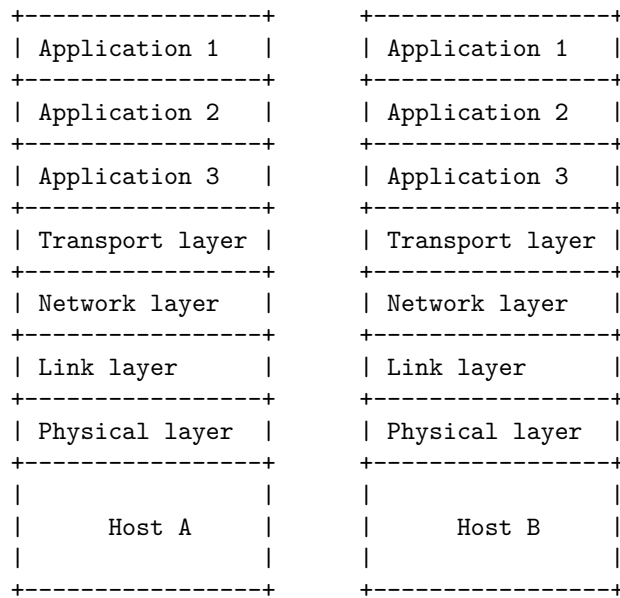
- TCP stands for Transmission Control Protocol .
- It is a transport layer protocol that facilitates the transmission of packets from source to destination .
- It is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network .
- TCP is a reliable protocol as it follows the flow and error control mechanism. It also supports the acknowledgment mechanism, which checks the state and sound arrival of the data .
- TCP uses a three-way handshake to establish the connection between the sender and the receiver. The three steps are: SYN, SYN-ACK, and ACK .
- TCP uses a four-way handshake to close the connection between the sender and the receiver. The four steps are: FIN, ACK, FIN, ACK .
- TCP provides various features such as congestion control, flow control, sequencing, retransmission, and multiplexing .
- TCP is suitable for applications that require reliable and ordered delivery of data, such as web browsing, email, file transfer, etc .

Multiplexing in transport layer

- Multiplexing is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver.
- Multiplexing in transport layer means extending the host-to-host delivery service provided by the network layer to a process-to-process delivery service for applications running on the hosts.
- Multiplexing in transport layer requires that sockets have unique identifiers, and each segment have special fields that indicate the sockets to which the segment is to be delivered.
- There are two types of multiplexing in transport layer: connectionless multiplexing and connection-oriented multiplexing.
- Connectionless multiplexing uses the User Datagram Protocol (UDP) to deliver data to the destination socket. UDP segments have two fields: source port number and destination port number, which identify the sockets at the sender and receiver respectively.
- Connection-oriented multiplexing uses the Transmission Control Protocol

(TCP) to deliver data to the destination socket. TCP segments have four fields: source port number, destination port number, source IP address and destination IP address, which identify the sockets at the sender and receiver respectively.

- Multiplexing in transport layer allows multiple applications to share the network resources and communicate with each other simultaneously.
- Multiplexing in transport layer is illustrated in the following diagram:



Connection management in transport layer

- The transport layer is responsible for creating and managing the end-to-end connections between hosts for data transmission.
- The transport layer uses two main protocols: TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).
- TCP is a reliable, connection-oriented protocol that uses a three-way handshake to establish a connection between two hosts. TCP ensures that the data is delivered in order and without errors, and provides flow control and congestion control mechanisms.
- UDP is an unreliable, connectionless protocol that does not guarantee the delivery, order, or integrity of the data. UDP is faster and simpler than TCP, and is used for applications that do not require reliability, such as streaming media or online games.
- Connection management involves three phases: connection establishment, data transfer, and connection termination.
- Connection establishment is the process of initiating a connection between two hosts. In TCP, this is done by exchanging SYN, SYN-ACK, and ACK

messages. In UDP, there is no connection establishment, and the hosts can start sending data packets immediately.

- Data transfer is the process of sending and receiving data between two hosts. In TCP, this is done by using sequence numbers, acknowledgments, and sliding windows to ensure reliable and ordered delivery of data. In UDP, there is no data transfer protocol, and the hosts simply send and receive datagrams without any feedback or control.
- Connection termination is the process of closing a connection between two hosts. In TCP, this is done by exchanging FIN, FIN-ACK, and ACK messages. In UDP, there is no connection termination, and the hosts can stop sending data at any time.

Flow control in transport layer

- Flow control is the mechanism that regulates the amount and rate of data transmission between two nodes in a network .
- Flow control is needed in transport layer to prevent data loss, buffer overflow or underflow, and congestion due to mismatched speeds or capacities of the sender and receiver .
- Flow control in transport layer is different from flow control in data link layer, as the former operates end-to-end across multiple links, while the latter operates locally over a single link.
- Flow control in transport layer can be implemented using various techniques, such as:
 - Stop-and-wait: The sender sends one packet and waits for an acknowledgment from the receiver before sending the next packet.
 - Sliding window: The sender and receiver agree on a window size, which is the number of packets that can be sent without waiting for an acknowledgment. The sender maintains a send window, which is the range of packets that are allowed to be sent. The receiver maintains a receive window, which is the range of packets that are allowed to be received. The sender and receiver update their windows based on the acknowledgments and feedback messages they exchange.
 - Flow control using TCP: The transport layer protocol TCP uses a combination of sliding window and congestion control techniques to achieve reliable and efficient data transmission. TCP adjusts the window size dynamically based on the network conditions and the receiver's buffer availability. TCP also uses acknowledgments, timers, sequence numbers, and checksums to ensure error-free and in-order delivery of packets .

Transmission in transport layer

- The transport layer is the fourth layer of the OSI model and the third layer of the TCP/IP model. It provides end-to-end communication services for applications, such as reliable data transfer, flow control, congestion control,

and error detection and correction.

- The transport layer uses protocols such as TCP and UDP to segment the data from the application layer into smaller units called segments or datagrams, and to add headers that contain information such as source and destination port numbers, sequence numbers, and checksums.
- The transport layer is responsible for ensuring that the data is transmitted reliably and efficiently across the network. It uses various mechanisms to achieve this, such as:
 - Acknowledgements: The receiver sends back a message to the sender to confirm the receipt of a segment or datagram. This helps the sender to keep track of the progress of the data transfer and to detect any lost or corrupted packets.
 - Retransmission: The sender re-sends a segment or datagram that has not been acknowledged by the receiver within a certain time interval. This helps to recover from packet loss and to ensure that the data is delivered completely and correctly.
 - Flow control: The sender adjusts the rate of data transmission according to the feedback from the receiver and the capacity of the network. This helps to avoid overwhelming the receiver or the network with too much data and to maintain a smooth and efficient data flow.
 - Congestion control: The sender monitors the level of congestion in the network and adapts the data transmission accordingly. This helps to prevent or reduce the congestion and to improve the network performance and quality of service.
 - Error detection and correction: The sender and the receiver use checksums or other methods to verify the integrity of the data and to detect any errors or corruption. If an error is detected, the sender or the receiver can request a retransmission or correct the error using techniques such as parity bits or error-correcting codes.
- The transport layer also provides other functions, such as:
 - Multiplexing and demultiplexing: The transport layer uses port numbers to identify different applications or processes on the same host and to deliver the data to the correct destination. This allows multiple applications or processes to share the same network connection and to communicate simultaneously.
 - Connection-oriented and connectionless services: The transport layer can provide different types of services depending on the protocol used and the requirements of the application. TCP provides a connection-oriented service, which means that it establishes a logical connection between the sender and the receiver before transmitting the data and maintains the connection until the data transfer is complete. UDP provides a connectionless service, which means that it does not establish or maintain any connection and simply sends the data without any guarantee of delivery or reliability.
 - Quality of service: The transport layer can support different levels of

quality of service (QoS) for different applications or processes, such as delay, bandwidth, jitter, and reliability. The transport layer can use various methods to achieve QoS, such as priority, reservation, or differentiation.

Window management in transport layer

- The transport layer is the layer in the Internet Protocol suite that provides end-to-end communication services for applications.
- The transport layer uses protocols such as Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) to establish connections, segment data, and ensure reliable delivery of packets.
- Window management is a technique used by the transport layer to control the flow of packets between two hosts and avoid congestion or data loss.
- Window management involves using a sliding window protocol, which is a method of sending and receiving data in fixed-sized frames or segments.
- A sliding window protocol works by maintaining a window size, which is the number of frames that can be sent or received at a time, and a window position, which is the sequence number of the first or last frame in the window.
- The sender and receiver exchange window size and position information using acknowledgment (ACK) and negative acknowledgment (NAK) messages, and adjust their windows accordingly.
- The sender can only send frames that are within its window, and the receiver can only accept frames that are within its window.
- The sender and receiver slide their windows forward when they receive an ACK or a NAK, respectively, indicating that a frame has been successfully received or lost.
- The window size can vary dynamically depending on the network conditions, such as the available bandwidth, the round-trip time, and the buffer occupancy.
- Window management can improve the efficiency and reliability of data transmission by avoiding buffer overflow, packet loss, and retransmission.

TCP Congestion Control in Transport Layer

- TCP congestion control is a mechanism that aims to regulate the amount of data that a sender can inject into the network, based on the network capacity and the feedback from the receiver .
- TCP congestion control consists of three main phases: slow start, congestion avoidance, and congestion detection .
- Slow start phase: The sender starts with a small congestion window (cwnd) and increases it exponentially for every acknowledgment (ACK) received, until it reaches a threshold value (ssthresh) or a packet loss occurs .
- Congestion avoidance phase: After reaching the ssthresh, the sender increases the cwnd linearly for every round-trip time (RTT), until a packet

loss occurs .

- Congestion detection phase: The sender detects a packet loss by either a timeout or a duplicate ACK. Depending on the detection method, the sender either reduces the cwnd by half (fast recovery) or sets it to one segment (slow start) and updates the ssthresh accordingly .
- TCP congestion control algorithms vary in how they adjust the cwnd and the ssthresh in response to network events. Some of the common algorithms are: Reno, NewReno, Tahoe, SACK, Vegas, BIC, CUBIC, etc .
- Some TCP congestion control algorithms require additional fields in the TCP header to carry congestion information, such as XCP and MaxNet.

Quality of service in transport layer

- Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity.
- The transport layer is responsible for enhancing the QoS provided by the network layer by offering reliable, efficient and secure data transfer between application processes.
- The transport layer can provide QoS by using the following techniques:
 - Establishing transport connections by sending requests and specifying the required QoS parameters, such as throughput, delay, jitter, reliability, etc. The transport layer uses transport addresses, which are pairs of network addresses and port numbers, to identify the end points of a connection .
 - Using different transport protocols, such as TCP or UDP, to suit the needs of different applications. TCP provides a connection-oriented, reliable, and flow-controlled service, while UDP provides a connectionless, unreliable, and datagram-based service.
 - Implementing error control, flow control, and congestion control mechanisms to ensure the integrity, efficiency, and fairness of data transmission. Error control detects and corrects errors in the data packets, flow control regulates the rate of data transmission between the sender and the receiver, and congestion control prevents the network from being overloaded by too many packets.
 - Providing security features, such as encryption, authentication, and integrity, to protect the data from unauthorized access, modification, or tampering. The transport layer can use cryptographic algorithms, such as SSL or TLS, to secure the data exchange between the application processes.

Unit 5 - Application Layer in Computer Networks

- The application layer is the topmost layer of the TCP/IP model that provides the interface between the applications and the network.

- The application layer defines the protocols that enable different processes and applications to communicate and exchange data over the network.
- The application layer protocols are specific to the type and purpose of the applications, such as web browsing, email, file transfer, remote login, etc.
- Some of the common application layer protocols are:
 - **Hypertext Transfer Protocol (HTTP)**: It is the protocol that governs the communication between web browsers and web servers. It defines how the browsers request and receive web pages, images, videos, and other resources from the servers. It also defines how the servers respond to the requests and send the requested resources. HTTP uses the TCP protocol at the transport layer to ensure reliable and ordered delivery of data.
 - **Domain Name System (DNS)**: It is the protocol that maps the human-readable domain names (such as `www.google.com`) to the numerical IP addresses (such as `172.217.160.78`) that identify the network devices. It enables the users to access the websites and services by using the domain names instead of the IP addresses. DNS uses both TCP and UDP protocols at the transport layer, depending on the size and type of the queries and responses.
 - **Simple Mail Transfer Protocol (SMTP)**: It is the protocol that enables the sending and receiving of email messages over the network. It defines how the email clients and servers interact and exchange the email messages. It also defines the format and structure of the email messages, such as the header, body, attachments, etc. SMTP uses the TCP protocol at the transport layer to ensure reliable and ordered delivery of email messages.
 - **Post Office Protocol (POP)** and **Internet Message Access Protocol (IMAP)**: They are the protocols that enable the email clients to retrieve the email messages from the email servers. They differ in the way they handle the email messages on the server. POP downloads the email messages from the server to the client and deletes them from the server, while IMAP synchronizes the email messages between the server and the client and keeps them on the server. POP and IMAP use the TCP protocol at the transport layer to ensure reliable and ordered delivery of email messages.
 - **File Transfer Protocol (FTP)**: It is the protocol that enables the transfer of files between the network devices. It defines how the file clients and servers interact and exchange the files. It also defines the commands and responses for establishing the connection, authentication, file listing, file uploading, file downloading, etc. FTP uses two TCP connections at the transport layer, one for control and one for data, to ensure reliable and ordered delivery of files.
 - **Telnet** and **Secure Shell (SSH)**: They are the protocols that enable the remote login and access to the network devices. They differ in the

way they handle the security and encryption of the data. Telnet sends the data in plain text, while SSH encrypts the data using public-key cryptography. Telnet and SSH use the TCP protocol at the transport layer to ensure reliable and ordered delivery of data.

Domain Name System

- The Domain Name System (DNS) is a hierarchical and decentralized system that translates human-readable domain names (such as `www.example.com`) into numerical IP addresses (such as `192.0.2.1`) that identify the devices and services on the Internet.
- The DNS consists of four main components: domain names, name servers, resolvers, and resource records.
- Domain names are the alphanumeric labels that are used to refer to specific websites or services on the Internet. They are organized into a tree-like structure, with the root domain at the top, followed by top-level domains (TLDs), such as `.com`, `.org`, `.net`, etc., and then subdomains, such as `www`, `mail`, etc. Each domain name has a unique owner and administrator, who can delegate authority to subdomains or subordinates.
- Name servers are the devices that store and provide the mapping between domain names and IP addresses. They are organized into a distributed and hierarchical network, with the root name servers at the top, followed by authoritative name servers for each TLD, and then authoritative name servers for each subdomain. Name servers communicate with each other using the DNS protocol, which defines the format and semantics of the queries and responses.
- Resolvers are the devices that initiate and process the DNS queries on behalf of the end users or applications. They are typically part of the operating system or the network configuration of the user's device. Resolvers send DNS queries to the name servers, starting from the root name servers and following the referrals until they reach the authoritative name server that has the answer. Resolvers cache the answers for a specified time to reduce the network traffic and latency.
- Resource records are the data elements that store the information about the domain names and IP addresses. They are stored in name servers and transferred in DNS queries and responses. Each resource record has a type, a name, a value, and a time-to-live (TTL). The type specifies the kind of information that the record contains, such as A (address), NS (name server), MX (mail exchange), etc. The name specifies the domain name that the record applies to, such as `www.example.com`. The value specifies the data that the record provides, such as `192.0.2.1`, `ns1.example.com`, `mail.example.com`, etc. The TTL specifies how long the record can be cached by the resolvers before it expires and needs to be refreshed.

World Wide Web

- The World Wide Web (WWW) is a global information system that uses the Internet as its infrastructure.
- The WWW consists of interconnected documents, called web pages, that are accessed through web browsers.
- Web pages are identified by unique addresses, called Uniform Resource Locators (URLs), that specify their location and protocol.
- Web pages can contain text, images, audio, video, and other multimedia elements, as well as hyperlinks to other web pages or resources.
- Web pages are created using a markup language, called Hypertext Markup Language (HTML), that defines their structure and appearance.
- Web pages can also use scripting languages, such as JavaScript, to add interactivity and functionality to the web pages.
- Web pages are delivered to web browsers by web servers, which are computers that store and serve web pages over the Internet.
- Web servers use a protocol, called Hypertext Transfer Protocol (HTTP), to communicate with web browsers and exchange web pages and other data.
- The WWW was invented by Tim Berners-Lee in 1989 at CERN, the European Organization for Nuclear Research, as a way to share information among researchers.
- The WWW is one of the most popular and widely used applications of the Internet, and has revolutionized many fields, such as education, communication, entertainment, commerce, and science.

Hyper Text Transfer Protocol

- Hyper Text Transfer Protocol (HTTP) is a protocol that defines how web browsers and web servers communicate and exchange data over the internet.
- HTTP is based on a request-response model, where a client (such as a browser) sends a request to a server (such as a web server) and the server responds with a status code, headers, and optionally a body.
- HTTP requests and responses can contain different types of data, such as text, images, audio, video, etc. The data is formatted using a standard syntax called Hyper Text Markup Language (HTML), which can include links to other resources (such as stylesheets, scripts, or other web pages).
- HTTP uses a uniform resource identifier (URI) to identify and locate a resource on the web. A URI consists of a scheme (such as http or https), a host (such as www.example.com), a port (such as 80 or 443), a path (such as /index.html), and optionally a query (such as ?q=hello) and a fragment (such as #section1).
- HTTP supports different methods (also called verbs) to perform different actions on a resource. The most common methods are GET, POST, PUT, DELETE, HEAD, and OPTIONS. Each method has a specific meaning

and semantics, such as retrieving, creating, updating, deleting, or obtaining information about a resource.

- HTTP also supports different status codes to indicate the outcome of a request. The status codes are grouped into five categories: 1xx (informational), 2xx (success), 3xx (redirection), 4xx (client error), and 5xx (server error). Some of the most common status codes are 200 (OK), 301 (Moved Permanently), 404 (Not Found), and 500 (Internal Server Error).
- HTTP is a stateless protocol, which means that each request and response are independent and do not remember any previous interaction. However, HTTP can use cookies, headers, or other mechanisms to maintain state and session information between requests and responses.
- HTTP is an extensible protocol, which means that it can be modified and enhanced by adding new methods, headers, status codes, or features. For example, HTTP/1.1 introduced persistent connections, chunked encoding, and caching. HTTP/2 introduced binary framing, multiplexing, compression, and server push. HTTP/3 introduced the use of QUIC, a new transport protocol that improves performance and security.

Electronic mail in application layer

- Electronic mail (or email) is an application layer service that allows users to exchange messages and information over the internet.
- Email is one of the most popular and widely used services of the internet .
- Email has three major components: user agents, mail servers, and protocols.
- User agents are the software programs that users interact with to read, compose, and organize email messages. Examples of user agents are Outlook, Gmail, Thunderbird, etc.
- Mail servers are the computers that store and forward email messages between user agents and other mail servers. Each mail server has a unique domain name and an IP address. Examples of mail servers are smtp.gmail.com, mail.yahoo.com, etc.
- Protocols are the rules and standards that govern the communication and transfer of email messages between user agents and mail servers. There are several protocols involved in email, such as SMTP, POP3, IMAP, MIME, etc.
- SMTP (Simple Mail Transfer Protocol) is the main protocol for sending email messages from a user agent to a mail server or from one mail server to another . SMTP uses TCP as the transport layer protocol and port 25 as the default port number .
- POP3 (Post Office Protocol version 3) and IMAP (Internet Message Access Protocol) are the protocols for retrieving email messages from a mail server to a user agent . POP3 allows the user agent to download and delete the messages from the mail server, while IMAP allows the user agent to access and manipulate the messages on the mail server without downloading

them. POP3 uses TCP and port 110, while IMAP uses TCP and port 143.

- MIME (Multipurpose Internet Mail Extensions) is the protocol for encoding and attaching non-text data, such as images, audio, video, etc., to email messages . MIME defines the format and structure of the email messages, such as the header, the body, the boundary, etc.

File Transfer Protocol in Application Layer

- File Transfer Protocol (FTP) is an application layer protocol that is used to transfer the files between the local devices (PC, smartphone, etc.) to a server .
- It transfers both text and binary files over the Internet.
- It runs on the top of TCP, like HTTP .
- To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection .
- Control connection is used for sending commands and receiving replies between the client and the server .
- Data connection is used for transferring the actual files between the client and the server .
- Control connection remains open throughout the FTP session, while data connection is opened and closed for each file transfer .
- FTP uses port number 21 for control connection and port number 20 for data connection .
- FTP supports two modes of data transfer: active mode and passive mode .
- In active mode, the server initiates the data connection to the client after receiving the client's IP address and port number .
- In passive mode, the client initiates the data connection to the server after receiving the server's IP address and port number .
- Passive mode is preferred when the client is behind a firewall or a NAT device .
- FTP requires the user to authenticate with a username and a password before accessing the server's files .
- FTP uses plaintext (unencrypted) sign-in process, which makes it vulnerable to eavesdropping and hacking .
- FTP can be secured by using encryption protocols such as FTPS (FTP over SSL/TLS) or SFTP (SSH File Transfer Protocol) .

Remote login in application layer

- Remote login is a process that allows an authorized user to access and use the services of another computer over a network, as if the user were physically present at the remote computer.
- Remote login is an example of an application layer service, which is the highest layer in the network protocol stack. The application layer provides

the interface between the user and the network, and defines the rules and formats for data exchange.

- Remote login can be implemented using different protocols, such as Telnet, SSH, Rlogin, etc. These protocols operate on the client-server model, where the client initiates a connection request to the server, and the server responds by authenticating the client and opening a session.
- Remote login can be used for various purposes, such as:
 - Remote administration: The user can perform system maintenance, configuration, troubleshooting, etc. on the remote computer.
 - Remote access: The user can access files, applications, databases, etc. on the remote computer.
 - Remote collaboration: The user can share the screen, keyboard, mouse, etc. of the remote computer with other users, and work together on a common task.
- Remote login has some advantages and disadvantages, such as:
 - Advantages:
 - * It saves time and resources, as the user does not need to physically travel to the remote location.
 - * It increases productivity and efficiency, as the user can work from anywhere and anytime.
 - * It enhances security and privacy, as the user can encrypt the data and use passwords to protect the connection.
 - Disadvantages:
 - * It requires a reliable and fast network connection, as any delay or interruption can affect the performance and quality of the service.
 - * It poses some risks and challenges, such as hacking, malware, unauthorized access, etc. that can compromise the security and integrity of the data and the system.
 - * It reduces the human interaction and socialization, as the user may feel isolated and disconnected from the remote environment.

Network management in application layer

- The application layer is the topmost layer in the Open System Interconnection (OSI) model, which defines how different devices communicate over a network.
- The application layer provides several ways for manipulating the data (information) which actually enables any type of user to access network with ease.
- The application layer handles network services, such as file and printing, name resolution, and redirector services. Name resolution is the process of mapping an IP address to a human-readable name.
- The application layer is where users interact with the network, download information and send data. The application layer also supports various protocols, such as HTTP, FTP, SMTP, DNS, etc., that enable different

applications to communicate over the network.

- Network management is the process of configuring, monitoring, and managing the performance of a network. Network management also involves ensuring network resources, such as hardware, storage, memory, bandwidth, data and processing power, are made readily accessible to users and services as efficiently and securely as possible.
- Network management can be performed at different layers of the OSI model, depending on the scope and complexity of the network. However, the application layer is the most visible and user-oriented layer of network management, as it deals with the quality and availability of network services and applications.
- Some of the benefits of network management at the application layer are:
 - It can help optimize the performance and reliability of network applications and services, such as web browsing, email, file transfer, etc.
 - It can help detect and troubleshoot network problems, such as latency, congestion, packet loss, etc., that affect the user experience and satisfaction.
 - It can help monitor and enforce network policies, such as security, access control, bandwidth allocation, etc., that ensure the network is used in a proper and efficient way.
 - It can help automate and streamline network operations, such as configuration, backup, recovery, etc., that reduce the workload and errors of network administrators.

Data compression in application layer

- Data compression is the process of reducing the number of bits required to represent a given data.
- Data compression can be performed at different layers of the OSI model, but it is most commonly done at the presentation layer or the application layer.
- The presentation layer is responsible for formatting, encoding, and transforming the data that is exchanged between the application layer and the network layer. It can also perform data compression, encryption, and translation functions to optimize the data transmission and storage.
- The application layer is the highest layer of the OSI model that provides the interface and services for the user applications. It can also perform data compression, encryption, and authentication functions to enhance the security and efficiency of the data communication.
- Some examples of application layer protocols that use data compression are:
 - File Transfer Protocol (FTP): It can compress the files before sending them over the network to reduce the bandwidth usage and the transfer time.
 - Simple Mail Transfer Protocol (SMTP): It can compress the email

messages and attachments before sending them to the recipients to save the disk space and the network resources.

- Domain Name System (DNS): It can compress the domain names and the IP addresses before resolving them to speed up the name resolution process.
- Data compression can have various benefits, such as:
 - Saving the network bandwidth and the disk space by reducing the size of the data.
 - Improving the performance and the speed of the data transmission and processing by reducing the latency and the overhead.
 - Enhancing the security and the privacy of the data by making it harder to read and modify by unauthorized parties.

Cryptography in application layer

- Cryptography is the science of securing data by transforming it into unintelligible form using algorithms and keys.
- Application layer encryption is a type of cryptography that encrypts data at the application level, rather than at the disk, file, or database level .
- Application layer encryption can increase security by reducing the number of potential attack vectors, as only authorized parties can access the encrypted data .
- Application layer encryption can also enable end-to-end encryption, which means that the data is encrypted from the sender to the receiver, without any intermediaries being able to decrypt it.
- Application layer encryption can be implemented using various techniques, such as symmetric encryption, asymmetric encryption, or hybrid encryption, depending on the use case and the security requirements.
- Application layer encryption can be used for various purposes, such as protecting sensitive data, ensuring data integrity, authenticating users, or enabling secure communication.
- Application layer encryption can be applied to different types of applications, such as native apps, browser apps, web apps, or mobile apps, using different tools and frameworks .

Basic concepts of cryptography in application layer

Cryptography is the science of securing communications from unauthorized parties. It involves the use of mathematical techniques to transform plain text into cipher text, which is unintelligible, and vice versa. Cryptography can provide confidentiality, integrity and authenticity to the data transmitted or stored in applications.

Some of the basic concepts of cryptography in application layer are:

- **Symmetric key cryptography:** This is a type of cryptography where the same key is used to encrypt and decrypt the data. The key must

be shared securely between the communicating parties. Symmetric key cryptography is fast and efficient, but it suffers from the key distribution problem. Examples of symmetric key algorithms are AES, DES, RC4, etc.

- **Asymmetric key cryptography:** This is a type of cryptography where a pair of keys is used: a public key and a private key. The public key can be shared openly, while the private key must be kept secret. The public key is used to encrypt the data, and the private key is used to decrypt it. Alternatively, the private key can be used to sign the data, and the public key can be used to verify it. Asymmetric key cryptography solves the key distribution problem of symmetric key cryptography, but it is slower and more complex. Examples of asymmetric key algorithms are RSA, ECC, DSA, etc.
 - **Hash functions:** These are mathematical functions that map any input to a fixed-length output, called a hash or a digest. Hash functions are one-way, meaning that it is easy to compute the hash from the input, but hard to find the input from the hash. Hash functions can be used to ensure the integrity of the data, by comparing the hash of the original data with the hash of the received data. Examples of hash functions are SHA, MD5, etc.
 - **Digital signatures:** These are a way of authenticating the sender and the data of a message, using asymmetric key cryptography. A digital signature is generated by applying a hash function to the data, and then encrypting the hash with the sender's private key. The receiver can verify the signature by decrypting it with the sender's public key, and comparing the hash with the hash of the data. Digital signatures can provide non-repudiation, meaning that the sender cannot deny sending the message. Examples of digital signature algorithms are RSA, DSA, ECDSA, etc.
 - **Encryption protocols:** These are sets of rules and procedures that define how cryptography is used to secure the communication between applications. Encryption protocols can specify the type of encryption algorithm, the key exchange method, the authentication mechanism, the error handling, etc. Examples of encryption protocols are SSL/TLS, SSH, IPsec, etc.
- : <https://resources.infosecinstitute.com/topic/how-is-cryptography-used-in-applications/>
<https://resources.infosecinstitute.com/topic/basics-of-cryptography-the-practical-application-and-use-of-cryptography/>
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